Trajectory Rewards Are Not Token Credit: Advantage Estimator Fidelity for Long-Horizon Post-Training

Limes Labs Research Workstream

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Abstract

Critic-free reinforcement learning, especially Group Relative Policy Optimization (GRPO), is attractive for reasoning-model post-training because it removes the learned value model and reduces memory pressure. This paper asks where that trade changes in long-horizon, variable-length, tool-using trajectories. Our working hypothesis is conditional: critic-based PPO is a better candidate when token- or state-level temporal credit is observable and covered, while GRPO remains compelling when group reward contrast is reliable and value modeling is weak or too expensive. OPD and OPSD are treated as complementary on-policy distillation objectives. The paper contributes a variance-reduction/credit-assignment taxonomy, a CPU-only estimator-fidelity testbed with known oracle advantages, and an exploratory coverage-gated credit estimator that falls back to group-relative credit when critic coverage is weak. Across a 20-seed, 18-case matrix, critic-style temporal-difference advantages have higher mean oracle-advantage correlation in 17 cases; a confidence-interval reading gives 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample. Closed-loop tabular policy training shows the same ordering direction but smaller downstream gaps. The result is mechanism evidence, not a frontier-model leaderboard claim.

Keywords. post-training; reinforcement learning from verifiable rewards; PPO; GRPO; on-policy distillation; credit assignment; long-horizon agents.

Contents

1 Introduction

Long-horizon post-training for language-model agents is not merely long-context language modeling. A trajectory may include partial plans, tool calls, retrievals, edits, failed attempts, retries, compacted summaries, and final verifier feedback. In such traces, the question "did the rollout succeed?" is not the same as "which token or action helped?" A scalar terminal reward can be useful for ranking responses, but it can also broadcast the same update over helpful subgoals, harmful detours, and no-op padding.

This paper is a public Limes Labs workstream on the method question: *for variable-length, multi-step, tool-heavy trajectories, when does a critic/value-model setup recover useful temporal credit that a group-relative setup discards or blurs?* We deliberately avoid centering the work on one base model. Z.ai's GLM-5.2 report is discussed only as an industry case study: it describes compacted, variable-length long-horizon rollouts and a move from group-wise optimization to critic-based PPO with token-level advantages [26]. That is not independent causal evidence that PPO beats GRPO; it is a motivating example of the problem structure studied here.

Contributions. We make four contributions. First, we separate PPO, GRPO, OPD, and OPSD by objective, signal granularity, and extra-model cost. Second, we give a cost and failure-mode checklist for fair post-training comparisons. Third, we introduce a small deterministic toy environment with oracle advantages and compare group-relative and critic-style advantage estimators across a multi-seed matrix. Fourth, we state testable predictions for future neural and agentic experiments.

Scope discipline. The paper makes a narrow claim about credit-assignment mechanisms. It does not claim that PPO is universally better than GRPO, that OPD is weaker than RL, or that a finite-state toy predicts frontier-model behavior. We use the toy because the oracle advantage is known, which lets us ask a question that is usually hidden in language-model training: whether an estimator is aligned with the token-level causal structure of a trajectory. This distinction matters for public research. A method can improve a benchmark for reasons unrelated to the mechanism under discussion, and a method can be mechanistically promising while still losing under a particular compute budget, implementation, reward model, or data mixture.

Paper organization. Section 2 defines the methods and separates optimization from distillation. Section 3 reviews related work. Section 4 gives cost accounting. Sections 5–7 define the toy environment, experimental protocol, and results. Sections 8–10 interpret the mechanism evidence, compare method regimes, and state failure modes. The appendices provide implementation details, case design, full summary tables, and raw per-seed rows generated directly from the canonical JSON result artifact.

2 Definitions and Method Taxonomy

Let a policy produce a trajectory

$$\tau = (s_0, a_0, s_1, a_1, \ldots, s_T), \tag{1}$$

where a state may include the prompt, partial response, tool outputs, verifier state, files, memory, or compacted context. Let $R(\tau)$ be an outcome reward, possibly combined with per-token KL, process rewards, or unit-test feedback.

2.1 PPO with a critic

PPO optimizes a clipped policy-gradient surrogate [19]:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[\min\left(\rho_t(\theta) A_t, \operatorname{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) A_t \right) \right], \tag{2}$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid s_t)}. \tag{3}$$

In actor-critic use, a value model V_ϕ can estimate one-step temporal difference errors

$$\delta_t = \nabla_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \tag{4}$$

or generalized advantage estimates [18]:

$$A_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \delta_{t+\ell}. \tag{5}$$

The critic is valuable only if the value target is learnable, sufficiently covered, and cheaper than the credit it recovers.

2.2 GRPO

GRPO samples a group of completions for the same prompt and normalizes rewards inside the group [4, 20]. A simplified response-level advantage is

$$\widehat{A}_i^{\mathrm{GRPO}} = \frac{R_i - \mu_G}{\sigma_G + \epsilon}, \tag{6}$$

where μ_G and σ_G are group statistics. This critic-free design is memory-efficient and effective in many verifiable-reward settings. Its long-horizon risk is intra-rollout information loss: one scalar can be broadcast across tokens with very different causal roles.

2.3 OPD and OPSD

OPD trains a student on student-sampled trajectories while a teacher provides dense supervision on those visited states [1, 10]. A typical objective is

$$L^{\mathrm{OPD}} = \mathbb{E}_{x\sim D,\, y\sim\pi_s(\cdot|x)} \left[\sum_t D_{\mathrm{KL}}(\pi_T(\cdot \mid x, y_{<t}) \,\|\, \pi_s(\cdot \mid x, y_{<t})) \right]. \tag{7}$$

OPSD removes the separate teacher model by making one model play a privileged teacher role and an ordinary student role over student rollouts [27]. These are dense distillation objectives. They can be excellent post-training tools, but they are not the same objective as reward optimization.

2.4 Variance reduction versus credit assignment

The critic in actor–critic PPO does two jobs at once. It acts as a state-dependent baseline that can reduce policy-gradient variance, and it creates token- or state-indexed temporal credit. These are different mechanisms. A running global baseline can reduce variance without telling us which step helped; a structural group method can remain critic-free while recovering some step-level contrast; a sampled value method can estimate per-state credit without training a value network.

Table 1: Method taxonomy for long-horizon post-training.

Method	Objective	Granularity	Extra signal	Main risk
PPO + critic	reward/KL opt.	tok./state	value, reward, ref.	critic cost; value miscalibration
GRPO	group-normalized RL	resp./group	verifier; no critic	weak intra-rollout credit
OPD	teacher distillation	token logits	teacher model	teacher mismatch; teacher cost
OPSD	privileged self-distillation	token logits	privileged traces	leakage; truncation collapse

Table 2: Two separable axes: how a baseline is formed and where credit is assigned. "SPO" is overloaded; here it refers separately to Single-stream Policy Optimization and Segment Policy Optimization.

Baseline source	Trajectory-level credit	Step-/segment-level credit
None	REINFORCE	–
Global/running	REINFORCE++; Single-stream PO	add process rewards
Sibling/LOO	GRPO; RLOO; DAPO; Dr. GRPO	GiGPO; SALT; PRM+GRPO
Learned critic $V(s)$	usually unnecessary alone	PPO/GAE; ArCHer
Sampled value	–	VinePPO; OPPO; Segment PO
Redistribution	–	RUDDER-style decomposition

2.5 Coverage-gated credit

The counterexample in this paper suggests a simple adaptive estimator. A critic should not be used merely because a value model exists; it should be used where the value estimate is locally supported. Let $n_D(s_t)$ be a replay or batch coverage count for the critic state abstraction, k a minimum support threshold, $\hat{A}_t^{\text{TD}}$ a critic temporal-difference advantage, and $\hat{A}_i^G$ the group-relative scalar for trajectory i. Define

$$m_t = \mathbf{1}\{n_D(s_t) \geq k\}, \qquad \hat{A}_t^{\text{CG}} = m_t \hat{A}_t^{\text{TD}} + (1 - m_t)\hat{A}_i^G. \tag{8}$$

This coverage-gated estimator is intentionally modest. It does not solve critic blindness: if the state abstraction hides the causal feature, high coverage can still be misleading. Its purpose is to make critic readiness an explicit experimental variable and to preserve group-relative signal where the critic is locally undercovered.

3 Related Work

PPO and actor–critic post-training. PPO was introduced as a simpler clipped-objective alternative to trust-region policy optimization [19]. In RLHF-style language-model training, actor–critic optimization is usually wrapped with KL penalties, reference policies, reward models, old-logprob storage, reward normalization, and sequence-level preference signals [16, 21]. The relevant point for this work is not one historical RLHF recipe, but the availability of a state- or token-indexed value estimate. A critic can turn delayed outcome feedback into temporally structured advantages. That can help when rollouts contain subgoals and detours, but it also creates a second model that can drift, overfit, or assign confident false credit.

GRPO and RL with verifiable rewards. DeepSeekMath introduced GRPO as a critic-free method for mathematical reasoning, replacing the learned value model with group-relative reward normalization [20]. DeepSeek-R1 then made GRPO-style RL a central public reference point for reasoning-model training [4]. Subsequent work has analyzed the effective loss, success amplification, response-length effects, dynamic sampling, clipping details, and statistical structure of GRPO-like gradients [13, 15, 17, 24, 28]. The practical attraction is real: GRPO removes critic memory and critic training, which can be decisive at large scale. The risk studied here is that a response-level group scalar can be information-wasteful inside long, heterogeneous rollouts.

Baselines, sampled values, and step credit. Recent work shows that the design space is broader than PPO versus GRPO. REINFORCE-style and global-normalization methods reduce critic cost without necessarily solving intra-rollout credit [2, 7, 23]. VinePPO estimates values by branching Monte Carlo rollouts rather than fitting a value network [8]. ArCHer uses a hierarchical multi-turn value structure for language-model agents [29]. OPPO is a very recent token-credit proposal based on Bayesian value recursion and should be read as a promising taxonomy anchor rather than settled evidence [11]. RUDDER and Segment Policy Optimization connect the same question to delayed-reward redistribution and segment-level credit [3, 6].

Dense and structural step signals. Process supervision is another way to avoid a pure terminal broadcast. Stepwise verification and process reward models provide direct intermediate feedback [12, 22], while implicit process rewards try to recover denser signals from cheaper outcome supervision [25]. GiGPO and SALT are especially relevant because they preserve critic-free or group-based structure while constructing finer-grained comparisons across repeated states or trajectory graphs [5, 9]. These methods make the main point sharper: "critic-free" does not have to mean "trajectory-only," and "critic-based" does not automatically mean correct credit.

On-policy distillation. OPD and related generalized knowledge distillation methods train on student-visited states, reducing train/inference mismatch relative to purely offline teacher forcing [1, 10]. OPSD uses the same model in privileged teacher and ordinary student roles, aiming to capture dense reasoning supervision without a separate teacher model [27]. Recent OPD analyses emphasize teacher/student compatibility, teacher novelty, length inflation, and truncation collapse [10, 14]. These methods are close to the present problem because they also produce dense token-level signals on rollouts. They differ because their signal is a distillation target rather than a reward-improvement objective.

Long-horizon agentic training. Long-horizon agentic RL introduces issues that are often absent in short verifier tasks: tool-call state, file-system state, hidden evaluation artifacts, environment side effects, compaction, and anti-hacking infrastructure. The GLM-5.2 report is useful because it explicitly names compacted sub-traces, variable numbers of trainable traces per prompt, and a move from group-wise optimization to critic-based PPO with token-level advantages [26]. We treat this as a case study of a design pressure, not as proof of a causal performance mechanism.

4 Cost Accounting

Fair comparisons should charge all hidden inputs, not only final benchmark score. For PPO, the bill includes value-model parameters, value targets, critic forward passes, optimizer state, old-policy log probabilities, KL control, and rollout storage. For GRPO, the bill includes group size, discarded completions, low-dispersion groups, and verifier calls. For OPD/OPSD, it includes teacher or privileged-context compute, dense logits, trajectory length inflation, and truncation handling. For agentic coding RL, it also includes anti-hacking filters, online tool-call monitors, LLM judges, sandboxing, blocked-call handling, and dummy observations when invalid calls are intercepted.

The near-term reporting scorecard for this workstream is the reward change reported beside its resource vector, not a literal scalar objective:

$$\Delta\text{reward} \quad \text{given} \quad (\text{generated tokens, wall clock, extra model memory}), \tag{9}$$

reported alongside reproducible artifacts and negative results.

Policy and rollout cost. For PPO, each trainable token typically needs old-policy log probabilities, advantages, reference-policy information for KL control, and value-model targets. If rollouts are compacted into sub-traces, the implementation must decide whether every sub-trace is trainable, how truncation is handled, and how losses are normalized across very different lengths. GRPO avoids the critic but spends compute on group sampling. If many group members have identical verifier outcomes, the effective signal can be weak even when token generation cost is high.

Verifier, judge, and safety cost. In coding-agent RL, pass/fail rewards can be corrupted by shortcuts such as reading hidden tests, fetching target solutions, or exploiting benchmark leakage. Online anti-hacking filters, rule-based monitors, LLM judges, dummy observations for blocked calls, and sandboxing are not optional bookkeeping details. They are part of the training objective as experienced by the agent. Ignoring them makes method comparisons look cleaner than they really are.

Distillation cost. OPD and OPSD can look cheap if only student updates are counted. A fair bill also includes teacher logits or hidden states, privileged traces, context construction, rejection/truncation policies, and any additional sampling needed to keep the student on-policy. Dense token supervision can be worth this cost, but it should not be compared to RL methods as if the dense signal were free.

5 Toy Experiment

5.1 Environment

The toy environment creates variable-length trajectories with three action types: `help`, `harm`, and `wait`. `help` increases a hidden progress score, `harm` decreases it, and `wait` consumes a token without changing progress. A terminal verifier returns success when the final score reaches a prompt-specific threshold. Because the finite dynamics are known, we can compute an oracle state-action advantage:

$$A^\star(s_t, a_t) = -c + V^\star(s_{t+1}, h-1) - V^\star(s_t, h), \tag{10}$$

where c is token cost and h is remaining horizon.

5.2 Estimators and metrics

The group-relative estimator normalizes terminal returns inside each prompt group and broadcasts the resulting scalar to all tokens in the trajectory. The critic-style estimator fits a tabular value model from sampled returns-to-go and computes one-step temporal-difference advantages. We measure Pearson correlation with the oracle advantage, calibrated MSE after affine rescaling, raw MSE, sign accuracy, wait-token leakage, within-trajectory variance, zero-variance group fraction, and exact critic-state hit rate.

The variance-credit grid adds six non-oracle estimators on the same toy dynamics: raw REINFORCE returns, a global return baseline, sibling group normalization, leave-one-out group centering, a dense progress-reward proxy, a learned critic TD estimator, and a sampled Monte Carlo value estimator. The purpose is not to simulate every cited algorithm, but to test the mechanism classes in Table 2: baseline-only variance reduction versus step-level credit.

Two robustness audits test likely objections. The length-imbalance audit adds a length-adjusted group scalar, computed by normalizing return per token inside each prompt group, and sweeps maximum horizon. The token-cost audit reruns baseline and long-wait settings with per-token costs $c \in \{0, 0.01, 0.02, 0.05\}$, including the no-explicit-length-penalty case $c = 0$.

The closed-loop training audit uses the same toy dynamics but trains a tabular softmax policy with policy-gradient updates under matched generated-token budgets. This is still not full PPO or production GRPO: there is no neural policy, KL controller, minibatch optimizer, or clipped surrogate. It does test whether estimator-fidelity differences survive one downstream optimization loop. The compared signals are group total return, group return per token, critic TD from a replay value table, and coverage-gated credit.

The oracle-alignment matrix remains an estimator-fidelity experiment under controlled dynamics. The closed-loop audit is a downstream tabular-policy stress test, not a production PPO-vs-GRPO benchmark.

6 Experimental Protocol

Design principle. The experiment is designed around a known oracle. In real language-model RL, we usually observe a reward and a model update but not the true token-level advantage. Here the finite-state dynamics allow exact dynamic programming for $V^\star$, which turns the question into an estimator-fidelity problem. That lets us separate two claims that are often conflated: whether a method is cheap and whether its advantage signal points at the right tokens.

Case axes. The deep matrix varies five axes. The horizon axis tests whether scalar terminal broadcasts become less aligned as trajectories lengthen. The group-size axis tests whether more completions per prompt rescue GRPO-like normalization. The critic-budget axis tests coverage: a value table trained from too few groups should not be trusted. The observability axis removes or coarsens state features to simulate poor value-model inputs. The sparse-reward axis tests harder terminal feedback. Finally, the counterexample intentionally combines a blind critic and low coverage to create a regime where group-relative terminal outcomes should win.

Decision rule. The headline statistic is Pearson correlation with the oracle advantage. A case is a mean critic win when the mean critic correlation exceeds the mean group-relative correlation. Because one mean win can be tiny, we also report a 95% across-seed confidence interval for the delta. A "clear critic" case has positive delta after subtracting the interval; a "clear group" case has negative delta after adding the interval; otherwise the case is labeled a near tie. This is why the abstract says both "17 mean critic wins" and the more cautious "16 clear, 1 near tie, 1 clear group" reading.

Why not a closed-loop toy RL run? A closed-loop toy PPO/GRPO implementation is a useful next step, but it would mix estimator fidelity with optimizer settings, clipping schedules, policy parameterization, exploration, and KL control. The current paper asks a more basic question first: when the oracle credit is known, which estimator is closer to it? That answer should inform later closed-loop experiments rather than be hidden inside them.

7 Results

The canonical matrix uses 20 seeds and 18 fixed cases across horizon length, group size, critic budget, observability, sparse-reward regimes, and a designed counterexample. Critic-style TD has the higher mean correlation in 17 of 18 cases. A confidence-interval reading is more cautious: 16 cases are clearly critic-favorable, 1 is a near tie, and 1 is clearly group-favorable. The mean critic-minus-group correlation is 0.420. Table 3 reports representative cases; the full CSV and JSON are committed in `results/`.

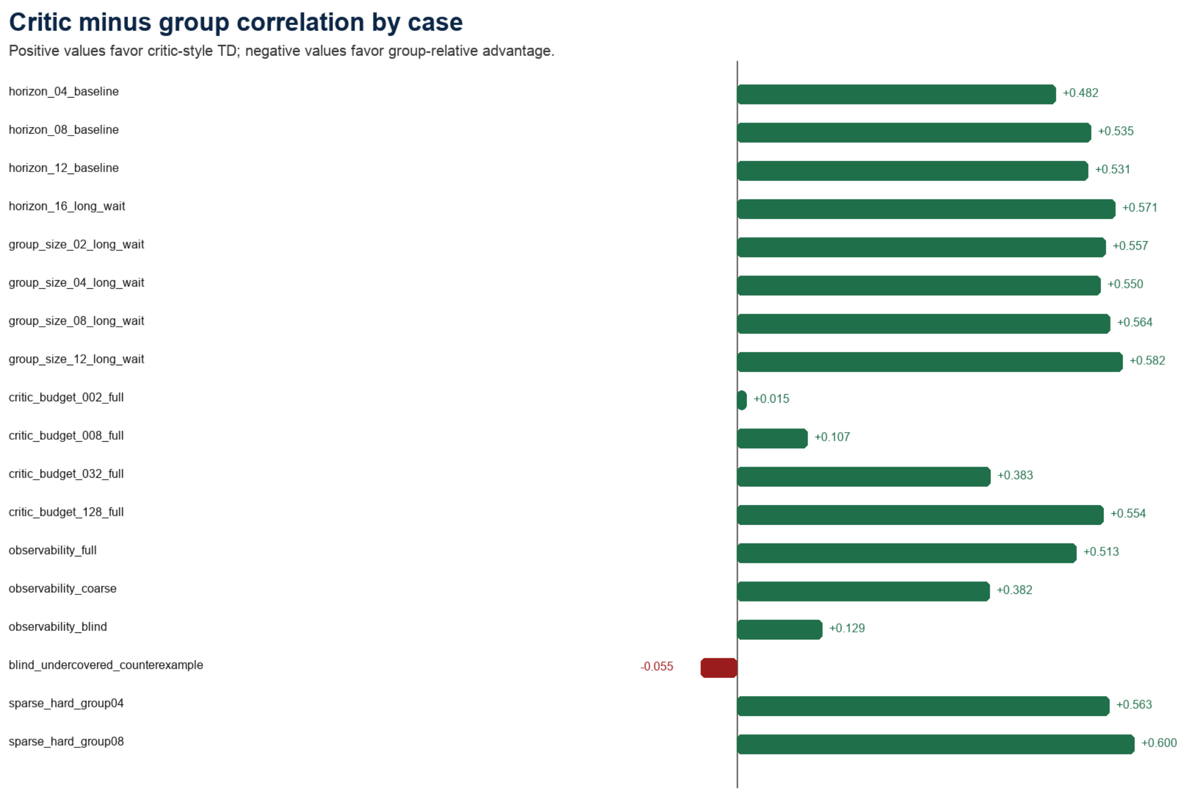

Figure 1: Mean critic-minus-group oracle-advantage correlation by case. Positive values favor critic-style temporal-difference advantages.

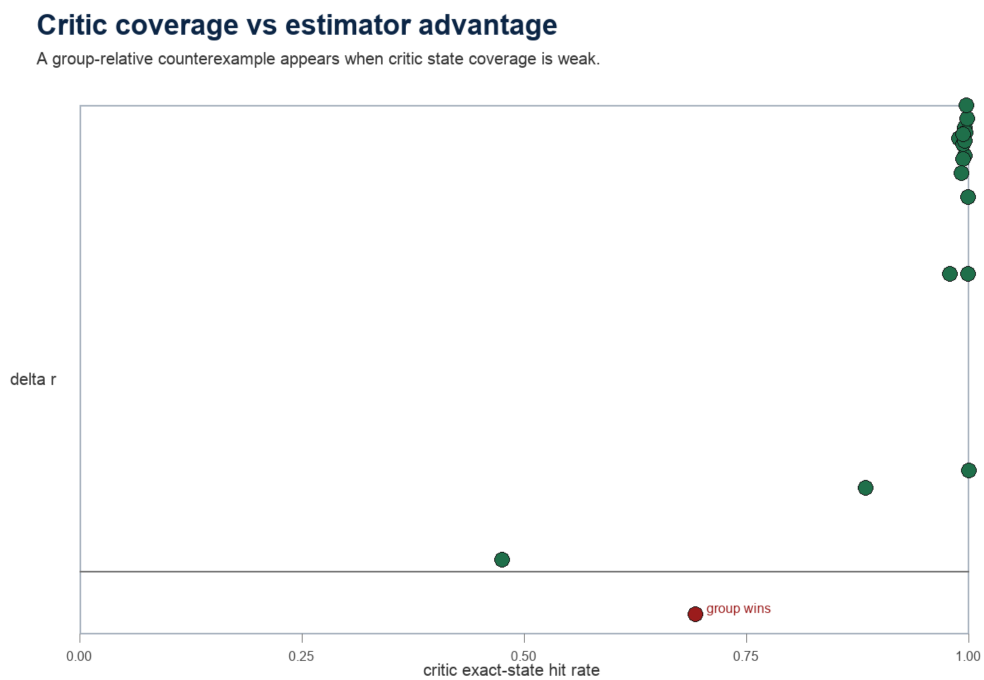

Figure 2: Critic state coverage versus estimator advantage. The group-relative counterexample occurs when the critic is blind and undercovered.

Table 3: Selected rows from the 20-seed deep matrix. Delta is critic r minus group r; CI is a 95% across-seed interval for delta.

Case	Group r	Critic r	Delta	CI	Reading
horizon_04_baseline	0.495	0.977	0.482	±0.011	critic clear
horizon_16_long_wait	0.293	0.865	0.571	±0.013	critic clear
critic_budget_002_full	0.352	0.367	0.015	±0.027	near tie
critic_budget_128_full	0.352	0.906	0.554	±0.017	critic clear
observability_blind	0.353	0.482	0.129	±0.017	critic clear
blind_undercovered_counterexample	0.510	0.455	-0.055	±0.015	group clear
sparse_hard_group08	0.310	0.910	0.600	±0.008	critic clear

Interpretation. Within this controlled estimator-fidelity toy, the evidence supports the PPO-favorable hypothesis when progress state is observable and critic coverage is adequate: critic-style advantages track token-level oracle credit more closely than response-level group scalars. The result also protects against dogma. In the coverage-limited case, the confidence interval crosses zero; in the blind and undercovered case, group-relative terminal outcome information is better than a poor value table.

Table 4: Variance-reduction versus credit-assignment grid on the long-wait toy case. Baseline-only methods can lower second moment, but they do not create within-trajectory credit variation. Abbreviations: Var. is variance reduction; Group z is sibling-group z-scoring; LOO is leave-one-out; Within is within-trajectory estimator variance.

Estimator	Var.	Credit	r	MSE	Wait	Within
REINFORCE	none	traj.	0.357	0.0208	1.157	0.000
Global	global	traj.	0.357	0.0208	0.875	0.000
Group z	group	traj.	0.319	0.0214	0.855	0.000
LOO	group	traj.	0.332	0.0212	0.799	0.000
Dense reward	dense	step	0.674	0.0130	0.000	0.416
Critic TD	critic	step	0.870	0.0058	0.356	0.031
Sampled MC	sampled	step	0.849	0.0067	0.563	0.034

Variance-credit decomposition. Table 4 makes the decomposition explicit. The global baseline reduces the second moment of the trajectory-level policy gradient signal relative to raw REINFORCE returns, but it leaves within-trajectory variance at zero because every token in a trajectory still receives the same scalar. Sibling group normalization is memory-light and critic-free, but in this long-wait setting it also remains a trajectory-level broadcast. The learned critic TD and sampled Monte Carlo value estimators both create step-level variation and substantially improve correlation with the oracle advantage. The dense progress-reward proxy sits between them: it gives a step signal, but because it lacks horizon and threshold information, it is not a full value estimate.

Table 5: Length-imbalance audit on the long-wait scenario. Len gap is the mean within-group max-minus-min trajectory length. Len-adj. is a group scalar based on return per token; Δ_G and Δ_L are critic r minus group-total and length-adjusted group r.

$H_{\max}$	Len gap	Wait	Group r	Len-adj. r	Critic r	Δ_G	Δ_L
4	1.775	0.336	0.488	0.500	0.981	0.492	0.481
8	4.851	0.431	0.389	0.389	0.940	0.551	0.551
12	7.725	0.495	0.330	0.328	0.890	0.561	0.562
16	10.696	0.539	0.293	0.289	0.849	0.556	0.560
20	13.439	0.573	0.266	0.260	0.811	0.545	0.551

Length-imbalance audit. Table 5 tests whether a simple length correction rescues trajectory-level grouping. It does not in this toy. As $H_{\max}$ increases from 4 to 20, mean within-group length range grows from 1.775 to 13.439 tokens and wait-token fraction grows from 0.336 to 0.573. The length-adjusted group scalar remains close to the total-return group scalar: at $H_{\max} = 20$, critic r is 0.811, group-total r is 0.266, and length-adjusted group r is 0.260. Length correction changes the rollout ranking signal, but it still broadcasts one number to all tokens.

Table 6: Token-cost sensitivity over 20 seeds. The $c = 0$ rows remove the explicit per-token penalty. CI is a 95% interval for critic-minus-group correlation; Wait_G and Wait_C are wait-token leakage ratios.

Scenario	Cost	Group r	Critic r	Δ	CI	Wait_G	Wait_C
base	0.00	0.336	0.882	0.546	0.010	0.930	0.465
base	0.01	0.332	0.882	0.550	0.010	0.937	0.465
base	0.02	0.335	0.882	0.546	0.010	0.928	0.465
base	0.05	0.339	0.882	0.543	0.010	0.902	0.465
long	0.00	0.335	0.888	0.552	0.015	0.871	0.371
long	0.01	0.327	0.888	0.561	0.016	0.911	0.371
long	0.02	0.330	0.888	0.557	0.016	0.898	0.371
long	0.05	0.334	0.888	0.554	0.016	0.868	0.371

Token-cost sensitivity. Table 6 checks whether the critic result is just a side effect of the per-token cost. The answer is no for these controlled dynamics. All eight scenario/cost rows have positive critic-minus-group correlation after subtracting the 95% interval. In the long-wait, zero-cost row, critic r is 0.888, group r is 0.335, and the delta is 0.552. Removing the explicit length penalty therefore does not remove the temporal-credit gap.

Table 7: Closed-loop tabular policy training under matched generated-token budgets. Default uses the standard replay budget; low cov. restricts replay and raises the coverage gate. Cov-gated is coverage-gated credit.

Setting	Method	Init R	Final R	ΔR	Success	Critic frac.
default	Group	0.515	0.785	0.271	0.949	0.000
default	Len-group	0.515	0.782	0.267	0.945	0.000
default	Critic TD	0.515	0.810	0.296	0.974	0.998
default	Cov-gated	0.515	0.809	0.294	0.972	0.996
low cov.	Group	0.508	0.769	0.261	0.928	0.000
low cov.	Len-group	0.508	0.754	0.246	0.913	0.000
low cov.	Critic TD	0.508	0.780	0.272	0.940	0.945
low cov.	Cov-gated	0.508	0.777	0.269	0.937	0.644

Closed-loop training. Table 7 is the first downstream optimization check. The result is more conservative than the oracle-fidelity tables, as expected. In the default setting, group-total training improves mean return from 0.515 to 0.785, while critic TD reaches 0.810 and coverage-gated credit reaches 0.809. In the low-coverage stress setting, group total reaches 0.769, critic TD reaches 0.780, and coverage-gated credit reaches 0.777 while using critic credit on only 64.4% of final-batch steps. The adaptive estimator is therefore a candidate design pattern, not a claimed replacement for PPO or GRPO: it preserves most critic benefit while making critic use explicit and auditable.

8 Mechanism Analysis

Horizon length. The horizon cases show the central intuition. As trajectories lengthen and wait tokens accumulate, a response-level scalar is asked to explain increasingly heterogeneous token roles. The group-relative estimator can still rank complete rollouts, but that does not mean it assigns good token credit. The critic-style estimator retains access to progress state and remaining horizon, so it can separate helpful actions from wait actions even when the final reward is delayed.

Group size. Increasing group size improves the quality of group statistics only when the additional samples create meaningful reward dispersion. In the long-wait cases, larger groups help the group-relative estimator somewhat, but the broadcast problem remains: every token in a completion inherits the same normalized outcome. In settings where verifier outcomes are sparse or clustered, larger groups can spend many tokens for little additional advantage information.

Critic budget and coverage. The critic-budget axis is the strongest warning against a simplistic "critics are better" claim. With only two training groups, the critic has an exact-state hit rate below one half and the confidence interval crosses zero. With more coverage, the critic delta grows sharply. In a neural setting, the analogous quantity is not table hit rate but value-model calibration under the current policy distribution. A poorly covered critic should be treated as a risk, not a badge of sophistication.

Observability. The observability cases show that a critic can still extract some value from coarse state, but a blind critic loses much of the temporal structure. The toy blind critic is intentionally simple; real long-horizon agents can become effectively blind for subtler reasons, including lossy compaction, missing tool state, stale memory, hidden environment variables, or value-model inputs that do not include the causal feature needed for future return.

Counterexample. The counterexample is not an embarrassment; it is the scientific control. If a paper about PPO versus GRPO cannot state when GRPO should win, it is not a method paper. In the blind-undercovered case, terminal group outcomes carry more usable information than a weak value table. The resulting group win is small but clear under the confidence-interval rule.

9 Which Method Is Better?

The answer is conditional, and the condition is about information. PPO-style critic methods are more plausible to test when future return is predictable from prefix state, when rollouts contain mixed token roles, and when the critic has enough coverage to be calibrated. GRPO-style methods are more plausible to test when rewards are cheap, verifiable, and meaningfully dispersed within prompt groups, or when the critic would be blind, stale, or too expensive. OPD and OPSD are more plausible when the available dense signal is a teacher or privileged self-teacher rather than an external reward.

This suggests a hybrid research program. Use rejection sampling and GRPO-like RL to find successes when verifiers are reliable. Use value or process models where temporal credit is the bottleneck. Use OPD/OPSD to distill verified or teacher-improved behaviors into cheaper policies. Report all of these with generated-token, verifier, teacher, critic, and wall-clock costs. The strongest method is unlikely to be a single objective used everywhere; it is more likely a pipeline whose components are selected by the information available at each training stage.

10 Failure Modes and Testable Predictions

PPO/critic failures. Critics can overfit reward artifacts, become stale after policy drift, collapse on long sparse trajectories, or assign confident but wrong temporal credit.

GRPO failures. GRPO can suffer zero-variance groups, low-dispersion amplification, response length bias, scalar reward broadcast over harmful tokens, and group composition confounding prompt difficulty with policy quality [13, 15, 17, 24, 28].

OPD/OPSD failures. Teacher/student thinking patterns may be incompatible; the teacher may offer no new information; on-policy rollouts may inflate in length and truncate; and privileged self-teacher traces may leak solution structure [10, 14, 27].

Predictions. We predict that increasing horizon length and no-op padding will reduce the Pearson alignment between response-level group advantages and oracle or process labels. We predict that increasing group size will help GRPO mainly when it increases within-prompt reward dispersion. We predict that adding process rewards will narrow the PPO/GRPO credit gap, and that degrading critic observability will produce regimes where GRPO wins. We predict that OPD/OPSD will be most token-efficient when teacher signals are novel and rollout lengths are controlled. Sampled-value methods should approach critic-like credit when the branching budget is high enough, but their inference cost should dominate in long agentic traces unless branch reuse or state sharing is available.

11 Threats to Validity

Toy dynamics. The toy has a small finite state, a known oracle, and a tabular critic. This is deliberately unlike frontier post-training. The benefit is interpretability; the cost is that neural generalization, distribution shift, optimizer dynamics, and reward-model error are not measured.

Estimator versus optimizer. The experiment compares advantage estimators, not complete PPO and GRPO training loops. A full optimizer could amplify or dampen estimator differences through clipping, KL schedules, entropy, minibatching, and policy parameterization. This paper therefore supports a mechanism hypothesis, not a leaderboard conclusion.

Simplified GRPO baseline. The group-relative baseline is intentionally simple. Future comparisons should include leave-one-out baselines, length-normalized objectives, Dr. GRPO-style corrections, DAPO-style dynamic sampling, token-level process rewards, and low-dispersion safeguards. A stronger GRPO family member may narrow the gap in some regimes.

Simplified critic. The critic is tabular and trained from returns-to-go. Neural value models can generalize across states, but they can also hallucinate value structure. The toy's exact-state hit rate is a proxy for coverage, not a complete model of critic reliability.

External validity. The GLM-5.2 discussion is based on a public vendor report and local text snapshot, not independent reproduction of its training stack. It should be read as evidence that the design pressure exists in industry systems, not as proof that critic-based PPO caused a benchmark gain.

12 Reproducibility

The core experiment code has no external Python dependencies. The canonical checks are:

```
python3 -m unittest discover -s tests
./scripts/run_smoke.sh
python3 -m experiments.deep_matrix \
  --output-json results/deep_matrix_20seed.json \
  --output-csv results/deep_matrix_20seed.csv \
  --output-md results/deep_matrix_20seed.md \
  --figures-dir results/figures
python3 -m experiments.variance_credit_grid \
  --output-json results/variance_credit_grid_seed17.json \
  --output-md results/variance_credit_grid_seed17.md
python3 -m experiments.length_imbalance_audit \
  --output-json results/length_imbalance_audit_seedset.json \
  --output-md results/length_imbalance_audit_seedset.md
python3 -m experiments.token_cost_sensitivity \
  --output-json results/token_cost_sensitivity_20seed.json \
  --output-md results/token_cost_sensitivity_20seed.md
python3 -m experiments.closed_loop_credit_training \
  --output-json results/closed_loop_credit_training_10seed.json \
  --output-md results/closed_loop_credit_training_10seed.md
./scripts/build_latex_paper.sh
```

The LaTeX paper is built with Tectonic when available and emitted as the full paper PDF artifact under `public/`.

The coverage-gated closed-loop proposal is also wired into Limes AutoResearch. From the repository root, set `PYTHONPATH` to the sibling `limes-autoresearch/repo` checkout and run:

```
python3 -m autoresearch_limes run \
  autoresearch/closed_loop_credit_training_config.json \
  --ledger autoresearch/runs/closed_loop_ledger.jsonl
```

The committed result card records the promotion gate and the fact that the current run is a candidate, not a verified replay.

13 Limitations

The toy environment has a finite known state and a tabular critic. Real post-training adds neural approximation, KL controllers, reward-model errors, tool failures, distributed rollout systems, stale policies, optimizer details, and data contamination risks. The GRPO baseline is intentionally simple; the next benchmark should include leave-one-out baselines, length-normalized variants, DAPO/Dr. GRPO-style corrections, process rewards, and dynamic sampling before making stronger claims.

14 Roadmap for Limes Labs

nanoGPT scale. The next controlled step is a tiny neural policy where the true environment is still known but the critic must generalize. This can test whether tabular coverage results survive function approximation.

AutoResearch and agentic tasks. For agentic research workflows, the key artifact is a trace schema that records tool state, compaction boundaries, blocked actions, reward calls, and process labels. Without that schema, value modeling and group-relative normalization cannot be audited fairly. The AutoResearch spec in this repository is a first hook: it turns coverage-gated credit into a bounded, ledgered research question with explicit baselines and promotion criteria.

EuroBench and Parameter Golf. Benchmark integrations should report cost-normalized improvement, not only final score. Parameter Golf is especially useful for comparing memory multipliers: a critic may improve credit assignment but violate a deployment budget. EuroBench-style tasks can test whether long-horizon behavior transfers from toy state to real language and tool use.

15 Conclusion

For long heterogeneous trajectories, critic-based PPO is a strong candidate when value targets are learnable and adequately covered. GRPO remains attractive when group reward contrast is reliable and critic modeling is too costly or poorly observed. OPD and OPSD are complementary dense-supervision tools that can transfer behaviors on student-visited states; unless they are paired with reward-improvement filtering, they answer a different objective from reward optimization. The right public research program is therefore not "PPO always beats GRPO," but a set of falsifiable boundaries for when temporal credit is worth its cost.

A Formal Objective Notes

Broadcasted group advantages. In a response-level group-relative update, a completion-level reward statistic is often applied to every trainable token in that completion. Let $g_i = (R_i - \mu_G)/(\sigma_G + \epsilon)$ be the group-normalized scalar for completion i. The token update is then proportional to $g_i \nabla_\theta \log \pi_\theta(a_{i,t} \mid s_{i,t})$ for many t. This is efficient, but it implicitly assumes that all trainable tokens in the completion should move in the same reward direction. Long-horizon traces violate that assumption whenever success depends on a small subset of steps and the rest are neutral or harmful.

Temporal-difference advantages. A critic-style estimator replaces one response-level scalar with a local difference between predicted future return before and after an action. In the toy, this is a tabular one-step TD estimate. In language-model agents, the same idea may be implemented with value heads, separate value models, process reward models, or learned subgoal predictors. The central requirement is that the state representation contains enough information to make future return predictable.

Distillation losses. The OPD/OPSD losses in the main text are written as KL objectives because the teacher signal is a distribution over next tokens. Other implementations may use sequence-level filtering, hidden-state matching, preference-filtered teacher traces, or mixed supervised/RL objectives. These variants can be useful without changing the taxonomy: they are dense imitation or self-imitation signals rather than direct policy-gradient reward optimization.

B Toy Environment Details

State. The state contains prompt identity, hidden progress, remaining horizon, and the features visible to the critic variant. The full critic observes enough to distinguish progress states. The coarse critic sees compressed state. The blind critic intentionally loses the feature needed for reliable future-return prediction.

Actions. The action set is deliberately small. `help` increases progress, `harm` decreases it, and `wait` consumes budget. This small action space makes exact dynamic programming feasible and makes the meaning of credit assignment visible. A language-model trace has richer tokens, but the causal categories are recognizable: helpful substeps, harmful detours, and neutral filler.

Reward. The terminal verifier is sparse and thresholded. This makes the group-relative estimator plausible, because rollouts can be ranked by final success, but also creates the credit problem: the reward is delayed until after many actions.

Oracle advantage. The oracle advantage uses exact values under known dynamics. This is the scientific anchor of the paper. Without it, a method comparison could only say which estimator produced higher downstream score under one optimizer. With it, we can ask whether an estimator points in the same direction as the true token-level credit.

C Case Design

The 18 fixed cases are grouped by axis. Horizon cases increase trace length and wait-token exposure. Group-size cases vary the number of completions used for within-prompt normalization. Critic-budget cases vary how much data the value table sees before evaluation. Observability cases degrade the state available to the critic. Sparse-reward cases make terminal success harder. The counterexample combines blindness and low coverage.

Table 8: Axis-level summary generated from the canonical matrix.

Axis	Cases	Group r	Critic r	Delta	Clear critic	Near tie	Clear group
counterexample	1	0.510	0.455	-0.055	0	0	1
critic_budget	4	0.352	0.617	0.265	3	1	0
group_size	4	0.305	0.868	0.563	4	0	0
horizon	4	0.379	0.908	0.530	4	0	0
observability	3	0.353	0.695	0.342	3	0	0
sparse_reward	2	0.296	0.878	0.582	2	0	0

D Full Case Summary

Table 9 gives the full 18-case summary. It is generated from the canonical matrix JSON rather than hand copied.

Table 9: Full 20-seed case summary. Delta is critic r minus group r.

Case	Axis	Read	Group r	Critic r	Δr	95% CI	Hit	Wait
horizon 04 baseline	horizon	critic	0.495	0.977	0.482	±0.011	0.999	0.303
horizon 08 baseline	horizon	critic	0.390	0.925	0.535	±0.015	0.996	0.365
horizon 12 baseline	horizon	critic	0.335	0.866	0.531	±0.012	0.994	0.411
horizon 16 long wait	horizon	critic	0.293	0.865	0.571	±0.013	0.996	0.539
group 02 long wait	group	critic	0.213	0.771	0.557	±0.022	0.989	0.505
group 04 long wait	group	critic	0.301	0.851	0.550	±0.014	0.994	0.503
group 08 long wait	group	critic	0.346	0.910	0.564	±0.011	0.997	0.503
group 12 long wait	group	critic	0.358	0.940	0.582	±0.007	0.998	0.501
budget 002 full	budget	near tie	0.352	0.367	0.015	±0.027	0.476	0.388
budget 008 full	budget	critic	0.352	0.459	0.107	±0.027	0.884	0.388
budget 032 full	budget	critic	0.352	0.735	0.383	±0.021	0.979	0.388
budget 128 full	budget	critic	0.352	0.906	0.554	±0.017	0.996	0.388
obs full	obs	critic	0.352	0.864	0.513	±0.020	0.992	0.388
obs coarse	obs	critic	0.356	0.738	0.382	±0.015	0.999	0.450
obs blind	obs	critic	0.353	0.482	0.129	±0.017	1.000	0.444

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Table 9: Full 20-seed case summary, continued.

Case	Axis	Read	Group r	Critic r	Δr	95% CI	Hit	Wait
blind undercov.	ctr.	group	0.510	0.455	-0.055	±0.015	0.693	0.329
sparse group04	sparse	critic	0.283	0.846	0.563	±0.014	0.994	0.417
sparse group08	sparse	critic	0.310	0.910	0.600	±0.008	0.997	0.420

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E Raw Seed-Level Rows

Table 10 records every case-by-seed row used to compute the paper's aggregate claims. Publishing these rows makes the mean-win and confidence-interval interpretation auditable. The table is long by design: there are 18 cases and 20 seeds, for 360 rows.

Table 10: Raw case-by-seed estimator-fidelity rows from the deep matrix.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
horizon 04 baseline	11	0.514	0.974	0.460	0.998	0.422	0.290
horizon 04 baseline	29	0.480	0.976	0.495	0.998	0.391	0.312
horizon 04 baseline	47	0.477	0.982	0.505	1.000	0.396	0.286
horizon 04 baseline	83	0.522	0.975	0.452	1.000	0.458	0.331
horizon 04 baseline	131	0.471	0.971	0.500	1.000	0.391	0.297
horizon 04 baseline	173	0.505	0.971	0.466	1.000	0.422	0.296
horizon 04 baseline	211	0.480	0.986	0.506	1.000	0.422	0.310
horizon 04 baseline	257	0.503	0.980	0.477	1.000	0.427	0.305
horizon 04 baseline	307	0.493	0.985	0.492	1.000	0.438	0.317
horizon 04 baseline	359	0.456	0.981	0.525	1.000	0.422	0.295
horizon 04 baseline	401	0.526	0.981	0.456	1.000	0.438	0.298
horizon 04 baseline	463	0.491	0.981	0.490	1.000	0.417	0.309
horizon 04 baseline	509	0.494	0.983	0.489	1.000	0.385	0.294
horizon 04 baseline	571	0.463	0.969	0.506	0.997	0.396	0.330
horizon 04 baseline	631	0.471	0.977	0.506	0.997	0.474	0.265
horizon 04 baseline	701	0.499	0.973	0.475	1.000	0.448	0.304
horizon 04 baseline	761	0.492	0.945	0.453	0.998	0.443	0.299
horizon 04 baseline	823	0.548	0.974	0.427	1.000	0.464	0.312
horizon 04 baseline	887	0.510	0.985	0.475	1.000	0.422	0.301
horizon 04 baseline	947	0.508	0.985	0.477	1.000	0.448	0.304
horizon 08 baseline	11	0.410	0.898	0.488	0.995	0.542	0.361
horizon 08 baseline	29	0.370	0.917	0.548	0.998	0.531	0.382
horizon 08 baseline	47	0.391	0.900	0.509	0.996	0.542	0.395
horizon 08 baseline	83	0.365	0.923	0.558	0.996	0.490	0.369
horizon 08 baseline	131	0.404	0.890	0.486	0.992	0.552	0.362
horizon 08 baseline	173	0.395	0.924	0.529	0.999	0.542	0.373
horizon 08 baseline	211	0.402	0.919	0.517	0.994	0.547	0.347
horizon 08 baseline	257	0.401	0.928	0.527	0.997	0.552	0.373
horizon 08 baseline	307	0.386	0.945	0.559	1.000	0.583	0.364
horizon 08 baseline	359	0.339	0.957	0.618	1.000	0.526	0.376
horizon 08 baseline	401	0.410	0.900	0.491	0.986	0.516	0.334
horizon 08 baseline	463	0.374	0.925	0.551	0.998	0.589	0.380
horizon 08 baseline	509	0.390	0.919	0.529	0.998	0.599	0.369
horizon 08 baseline	571	0.376	0.973	0.596	0.999	0.568	0.347
horizon 08 baseline	631	0.379	0.947	0.568	1.000	0.510	0.341
horizon 08 baseline	701	0.400	0.925	0.526	0.996	0.583	0.404
horizon 08 baseline	761	0.380	0.916	0.536	0.996	0.547	0.350
horizon 08 baseline	823	0.420	0.913	0.494	0.990	0.578	0.357
horizon 08 baseline	887	0.408	0.937	0.530	0.995	0.604	0.364
horizon 08 baseline	947	0.405	0.945	0.540	1.000	0.500	0.358
horizon 12 baseline	11	0.314	0.868	0.554	0.999	0.609	0.415
horizon 12 baseline	29	0.348	0.919	0.571	0.995	0.589	0.400

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
horizon 12 baseline	47	0.321	0.870	0.549	0.997	0.677	0.417
horizon 12 baseline	83	0.346	0.869	0.524	0.994	0.573	0.396
horizon 12 baseline	131	0.339	0.878	0.540	0.994	0.604	0.397
horizon 12 baseline	173	0.357	0.836	0.479	0.994	0.630	0.416
horizon 12 baseline	211	0.337	0.860	0.523	0.996	0.667	0.390
horizon 12 baseline	257	0.346	0.893	0.547	0.997	0.656	0.437
horizon 12 baseline	307	0.324	0.877	0.554	0.992	0.667	0.412
horizon 12 baseline	359	0.349	0.846	0.497	0.995	0.599	0.397
horizon 12 baseline	401	0.349	0.870	0.521	0.993	0.630	0.412
horizon 12 baseline	463	0.348	0.842	0.494	0.997	0.656	0.416
horizon 12 baseline	509	0.330	0.887	0.557	0.992	0.661	0.425
horizon 12 baseline	571	0.326	0.840	0.514	0.986	0.661	0.424
horizon 12 baseline	631	0.342	0.847	0.505	0.990	0.604	0.392
horizon 12 baseline	701	0.316	0.895	0.578	0.996	0.651	0.445
horizon 12 baseline	761	0.349	0.884	0.536	0.990	0.625	0.405
horizon 12 baseline	823	0.341	0.847	0.507	0.990	0.573	0.399
horizon 12 baseline	887	0.314	0.852	0.538	0.998	0.646	0.402
horizon 12 baseline	947	0.314	0.844	0.530	0.993	0.620	0.421
horizon 16 long wait	11	0.302	0.862	0.560	0.998	0.667	0.513
horizon 16 long wait	29	0.303	0.884	0.581	0.998	0.667	0.536
horizon 16 long wait	47	0.266	0.866	0.601	0.998	0.718	0.538
horizon 16 long wait	83	0.319	0.885	0.565	0.996	0.620	0.529
horizon 16 long wait	131	0.292	0.871	0.579	0.995	0.671	0.542
horizon 16 long wait	173	0.297	0.869	0.572	0.997	0.606	0.552
horizon 16 long wait	211	0.292	0.875	0.583	0.996	0.718	0.546
horizon 16 long wait	257	0.299	0.865	0.566	0.991	0.694	0.532
horizon 16 long wait	307	0.264	0.909	0.645	0.998	0.671	0.539
horizon 16 long wait	359	0.287	0.847	0.560	0.994	0.741	0.546
horizon 16 long wait	401	0.291	0.809	0.517	0.993	0.718	0.539
horizon 16 long wait	463	0.299	0.860	0.561	0.997	0.662	0.552
horizon 16 long wait	509	0.306	0.875	0.568	0.995	0.662	0.545
horizon 16 long wait	571	0.297	0.826	0.529	0.994	0.657	0.547
horizon 16 long wait	631	0.305	0.840	0.534	0.995	0.667	0.517
horizon 16 long wait	701	0.291	0.845	0.554	0.996	0.713	0.527
horizon 16 long wait	761	0.316	0.878	0.562	0.997	0.736	0.562
horizon 16 long wait	823	0.287	0.866	0.579	0.995	0.657	0.532
horizon 16 long wait	887	0.272	0.895	0.623	0.998	0.713	0.539
horizon 16 long wait	947	0.279	0.868	0.589	0.998	0.694	0.545
group 02 long wait	11	0.174	0.665	0.492	0.968	0.641	0.507
group 02 long wait	29	0.227	0.774	0.547	0.991	0.719	0.496
group 02 long wait	47	0.223	0.792	0.569	0.994	0.656	0.495
group 02 long wait	83	0.245	0.709	0.464	0.987	0.719	0.487
group 02 long wait	131	0.189	0.758	0.569	0.998	0.625	0.514
group 02 long wait	173	0.212	0.794	0.582	0.995	0.734	0.499
group 02 long wait	211	0.216	0.773	0.557	0.987	0.578	0.489
group 02 long wait	257	0.236	0.775	0.539	0.986	0.641	0.515
group 02 long wait	307	0.187	0.811	0.624	0.995	0.672	0.533
group 02 long wait	359	0.212	0.699	0.487	0.980	0.625	0.453
group 02 long wait	401	0.195	0.720	0.525	0.996	0.672	0.528
group 02 long wait	463	0.184	0.805	0.621	0.993	0.609	0.496
group 02 long wait	509	0.223	0.754	0.532	0.979	0.656	0.515
group 02 long wait	571	0.247	0.878	0.631	0.986	0.656	0.536
group 02 long wait	631	0.214	0.777	0.563	0.998	0.641	0.492
group 02 long wait	701	0.263	0.748	0.485	0.978	0.609	0.488
group 02 long wait	761	0.225	0.783	0.558	0.998	0.672	0.490
group 02 long wait	823	0.207	0.795	0.588	0.994	0.688	0.498
group 02 long wait	887	0.203	0.774	0.571	0.987	0.734	0.537
group 02 long wait	947	0.187	0.827	0.641	0.991	0.703	0.542
group 04 long wait	11	0.273	0.788	0.515	0.991	0.680	0.488
group 04 long wait	29	0.305	0.861	0.557	0.998	0.734	0.495
group 04 long wait	47	0.265	0.794	0.529	0.988	0.656	0.481
group 04 long wait	83	0.300	0.874	0.574	1.000	0.664	0.503

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
group 04 long wait	131	0.323	0.867	0.544	0.998	0.625	0.524
group 04 long wait	173	0.333	0.796	0.464	0.985	0.656	0.465
group 04 long wait	211	0.291	0.846	0.555	0.999	0.609	0.507
group 04 long wait	257	0.308	0.875	0.567	0.996	0.664	0.533
group 04 long wait	307	0.331	0.879	0.548	0.991	0.641	0.495
group 04 long wait	359	0.289	0.859	0.570	0.993	0.609	0.471
group 04 long wait	401	0.298	0.909	0.611	0.999	0.656	0.544
group 04 long wait	463	0.293	0.880	0.587	0.995	0.570	0.487
group 04 long wait	509	0.300	0.825	0.525	0.992	0.617	0.516
group 04 long wait	571	0.321	0.879	0.558	0.988	0.688	0.514
group 04 long wait	631	0.303	0.841	0.538	0.998	0.680	0.485
group 04 long wait	701	0.291	0.876	0.585	0.997	0.688	0.517
group 04 long wait	761	0.296	0.871	0.575	1.000	0.688	0.491
group 04 long wait	823	0.296	0.857	0.561	0.982	0.664	0.508
group 04 long wait	887	0.282	0.799	0.517	0.995	0.672	0.508
group 04 long wait	947	0.330	0.847	0.517	0.997	0.633	0.516
group 08 long wait	11	0.337	0.903	0.566	0.997	0.656	0.506
group 08 long wait	29	0.345	0.915	0.569	0.998	0.652	0.493
group 08 long wait	47	0.301	0.907	0.607	0.998	0.680	0.506
group 08 long wait	83	0.356	0.931	0.575	0.997	0.625	0.460
group 08 long wait	131	0.373	0.931	0.558	0.996	0.637	0.522
group 08 long wait	173	0.353	0.865	0.512	0.996	0.633	0.497
group 08 long wait	211	0.357	0.927	0.570	0.998	0.621	0.503
group 08 long wait	257	0.344	0.938	0.595	0.998	0.617	0.505
group 08 long wait	307	0.333	0.925	0.592	0.998	0.664	0.511
group 08 long wait	359	0.350	0.891	0.541	0.995	0.660	0.505
group 08 long wait	401	0.343	0.924	0.581	0.998	0.660	0.515
group 08 long wait	463	0.340	0.900	0.560	0.999	0.609	0.518
group 08 long wait	509	0.333	0.925	0.592	0.996	0.691	0.503
group 08 long wait	571	0.333	0.867	0.534	0.994	0.672	0.526
group 08 long wait	631	0.366	0.904	0.538	0.997	0.609	0.517
group 08 long wait	701	0.342	0.899	0.556	0.998	0.668	0.505
group 08 long wait	761	0.343	0.945	0.602	0.998	0.586	0.494
group 08 long wait	823	0.346	0.887	0.541	0.989	0.617	0.484
group 08 long wait	887	0.369	0.914	0.545	0.997	0.613	0.484
group 08 long wait	947	0.350	0.900	0.550	0.996	0.660	0.513
group 12 long wait	11	0.365	0.924	0.559	0.999	0.646	0.496
group 12 long wait	29	0.367	0.935	0.568	0.999	0.633	0.491
group 12 long wait	47	0.338	0.958	0.620	0.999	0.659	0.511
group 12 long wait	83	0.372	0.939	0.568	0.999	0.674	0.498
group 12 long wait	131	0.353	0.958	0.605	1.000	0.638	0.531
group 12 long wait	173	0.355	0.948	0.593	0.998	0.690	0.490
group 12 long wait	211	0.351	0.948	0.596	0.998	0.617	0.488
group 12 long wait	257	0.352	0.940	0.588	0.999	0.607	0.511
group 12 long wait	307	0.367	0.943	0.576	1.000	0.654	0.502
group 12 long wait	359	0.348	0.938	0.590	0.998	0.667	0.493
group 12 long wait	401	0.363	0.937	0.574	0.997	0.674	0.502
group 12 long wait	463	0.356	0.919	0.564	0.997	0.646	0.520
group 12 long wait	509	0.352	0.942	0.590	0.996	0.635	0.490
group 12 long wait	571	0.368	0.928	0.560	0.996	0.628	0.511
group 12 long wait	631	0.371	0.937	0.566	0.998	0.656	0.507
group 12 long wait	701	0.367	0.945	0.578	1.000	0.693	0.513
group 12 long wait	761	0.344	0.934	0.590	0.998	0.659	0.500
group 12 long wait	823	0.348	0.947	0.599	0.997	0.617	0.486
group 12 long wait	887	0.356	0.942	0.585	0.998	0.635	0.496
group 12 long wait	947	0.363	0.942	0.579	0.999	0.656	0.494
budget 002 full	11	0.358	0.415	0.057	0.467	0.573	0.392
budget 002 full	29	0.395	0.405	0.009	0.476	0.620	0.400
budget 002 full	47	0.342	0.330	-0.012	0.536	0.609	0.385
budget 002 full	83	0.308	0.251	-0.056	0.439	0.630	0.409
budget 002 full	131	0.319	0.340	0.021	0.369	0.646	0.381
budget 002 full	173	0.328	0.441	0.113	0.514	0.562	0.381

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
budget 002 full	211	0.363	0.299	-0.064	0.501	0.630	0.398
budget 002 full	257	0.359	0.451	0.091	0.459	0.516	0.375
budget 002 full	307	0.357	0.267	-0.090	0.522	0.594	0.380
budget 002 full	359	0.313	0.414	0.101	0.376	0.568	0.364
budget 002 full	401	0.385	0.304	-0.081	0.506	0.573	0.374
budget 002 full	463	0.381	0.484	0.102	0.454	0.557	0.407
budget 002 full	509	0.329	0.358	0.029	0.489	0.604	0.381
budget 002 full	571	0.381	0.379	-0.003	0.432	0.599	0.397
budget 002 full	631	0.364	0.306	-0.058	0.470	0.562	0.388
budget 002 full	701	0.318	0.368	0.050	0.543	0.594	0.406
budget 002 full	761	0.329	0.375	0.046	0.503	0.651	0.413
budget 002 full	823	0.387	0.409	0.022	0.389	0.615	0.392
budget 002 full	887	0.377	0.368	-0.009	0.528	0.609	0.360
budget 002 full	947	0.342	0.383	0.040	0.539	0.599	0.372
budget 008 full	11	0.358	0.471	0.114	0.868	0.573	0.392
budget 008 full	29	0.395	0.536	0.141	0.873	0.620	0.400
budget 008 full	47	0.342	0.507	0.165	0.913	0.609	0.385
budget 008 full	83	0.308	0.470	0.163	0.856	0.630	0.409
budget 008 full	131	0.319	0.395	0.076	0.878	0.646	0.381
budget 008 full	173	0.328	0.567	0.239	0.885	0.562	0.381
budget 008 full	211	0.363	0.466	0.104	0.846	0.630	0.398
budget 008 full	257	0.359	0.517	0.158	0.917	0.516	0.375
budget 008 full	307	0.357	0.399	0.042	0.926	0.594	0.380
budget 008 full	359	0.313	0.530	0.217	0.872	0.568	0.364
budget 008 full	401	0.385	0.480	0.096	0.885	0.573	0.374
budget 008 full	463	0.381	0.455	0.074	0.958	0.557	0.407
budget 008 full	509	0.329	0.435	0.106	0.913	0.604	0.381
budget 008 full	571	0.381	0.465	0.084	0.875	0.599	0.397
budget 008 full	631	0.364	0.353	-0.012	0.828	0.562	0.388
budget 008 full	701	0.318	0.422	0.104	0.884	0.594	0.406
budget 008 full	761	0.329	0.348	0.020	0.847	0.651	0.413
budget 008 full	823	0.387	0.429	0.042	0.849	0.615	0.392
budget 008 full	887	0.377	0.501	0.124	0.903	0.609	0.360
budget 008 full	947	0.342	0.434	0.092	0.902	0.599	0.372
budget 032 full	11	0.358	0.715	0.357	0.975	0.573	0.392
budget 032 full	29	0.395	0.761	0.365	0.974	0.620	0.400
budget 032 full	47	0.342	0.749	0.407	0.987	0.609	0.385
budget 032 full	83	0.308	0.709	0.402	0.975	0.630	0.409
budget 032 full	131	0.319	0.715	0.396	0.974	0.646	0.381
budget 032 full	173	0.328	0.828	0.500	0.980	0.562	0.381
budget 032 full	211	0.363	0.667	0.305	0.971	0.630	0.398
budget 032 full	257	0.359	0.795	0.436	0.993	0.516	0.375
budget 032 full	307	0.357	0.708	0.351	0.983	0.594	0.380
budget 032 full	359	0.313	0.780	0.467	0.982	0.568	0.364
budget 032 full	401	0.385	0.691	0.306	0.973	0.573	0.374
budget 032 full	463	0.381	0.757	0.375	0.987	0.557	0.407
budget 032 full	509	0.329	0.672	0.343	0.981	0.604	0.381
budget 032 full	571	0.381	0.717	0.336	0.979	0.599	0.397
budget 032 full	631	0.364	0.720	0.356	0.961	0.562	0.388
budget 032 full	701	0.318	0.700	0.382	0.969	0.594	0.406
budget 032 full	761	0.329	0.700	0.372	0.973	0.651	0.413
budget 032 full	823	0.387	0.798	0.411	0.993	0.615	0.392
budget 032 full	887	0.377	0.771	0.394	0.985	0.609	0.360
budget 032 full	947	0.342	0.740	0.397	0.984	0.599	0.372
budget 128 full	11	0.358	0.907	0.549	0.994	0.573	0.392
budget 128 full	29	0.395	0.884	0.489	0.996	0.620	0.400
budget 128 full	47	0.342	0.909	0.567	0.997	0.609	0.385
budget 128 full	83	0.308	0.896	0.588	0.996	0.630	0.409
budget 128 full	131	0.319	0.899	0.580	0.992	0.646	0.381
budget 128 full	173	0.328	0.942	0.614	1.000	0.562	0.381
budget 128 full	211	0.363	0.873	0.510	0.993	0.630	0.398
budget 128 full	257	0.359	0.917	0.557	0.998	0.516	0.375

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
budget 128 full	307	0.357	0.880	0.523	0.993	0.594	0.380
budget 128 full	359	0.313	0.938	0.625	0.997	0.568	0.364
budget 128 full	401	0.385	0.905	0.521	0.992	0.573	0.374
budget 128 full	463	0.381	0.915	0.533	0.996	0.557	0.407
budget 128 full	509	0.329	0.923	0.594	0.993	0.604	0.381
budget 128 full	571	0.381	0.880	0.498	0.997	0.599	0.397
budget 128 full	631	0.364	0.881	0.516	0.987	0.562	0.388
budget 128 full	701	0.318	0.909	0.591	0.999	0.594	0.406
budget 128 full	761	0.329	0.897	0.568	0.997	0.651	0.413
budget 128 full	823	0.387	0.935	0.549	0.996	0.615	0.392
budget 128 full	887	0.377	0.914	0.537	0.998	0.609	0.360
budget 128 full	947	0.342	0.919	0.577	0.998	0.599	0.372
obs full	11	0.358	0.843	0.485	0.991	0.573	0.392
obs full	29	0.395	0.856	0.460	0.992	0.620	0.400
obs full	47	0.342	0.867	0.525	0.996	0.609	0.385
obs full	83	0.308	0.872	0.564	0.993	0.630	0.409
obs full	131	0.319	0.843	0.524	0.984	0.646	0.381
obs full	173	0.328	0.935	0.607	1.000	0.562	0.381
obs full	211	0.363	0.835	0.473	0.987	0.630	0.398
obs full	257	0.359	0.871	0.512	0.995	0.516	0.375
obs full	307	0.357	0.868	0.510	0.991	0.594	0.380
obs full	359	0.313	0.890	0.577	0.993	0.568	0.364
obs full	401	0.385	0.809	0.424	0.989	0.573	0.374
obs full	463	0.381	0.870	0.489	0.994	0.557	0.407
obs full	509	0.329	0.893	0.564	0.989	0.604	0.381
obs full	571	0.381	0.836	0.454	0.993	0.599	0.397
obs full	631	0.364	0.845	0.481	0.984	0.562	0.388
obs full	701	0.318	0.831	0.513	0.989	0.594	0.406
obs full	761	0.329	0.859	0.531	0.996	0.651	0.413
obs full	823	0.387	0.891	0.504	0.996	0.615	0.392
obs full	887	0.377	0.894	0.517	0.995	0.609	0.360
obs full	947	0.342	0.881	0.539	0.996	0.599	0.372
obs coarse	11	0.361	0.701	0.341	1.000	0.568	0.440
obs coarse	29	0.378	0.710	0.331	1.000	0.630	0.453
obs coarse	47	0.351	0.702	0.352	1.000	0.646	0.453
obs coarse	83	0.321	0.773	0.452	1.000	0.578	0.456
obs coarse	131	0.344	0.727	0.382	1.000	0.615	0.434
obs coarse	173	0.355	0.728	0.373	1.000	0.609	0.446
obs coarse	211	0.366	0.735	0.370	0.997	0.630	0.453
obs coarse	257	0.380	0.767	0.386	1.000	0.547	0.441
obs coarse	307	0.342	0.710	0.368	1.000	0.620	0.452
obs coarse	359	0.328	0.760	0.432	1.000	0.599	0.461
obs coarse	401	0.371	0.758	0.387	1.000	0.625	0.443
obs coarse	463	0.367	0.703	0.336	1.000	0.604	0.457
obs coarse	509	0.308	0.750	0.442	1.000	0.635	0.441
obs coarse	571	0.379	0.760	0.382	1.000	0.625	0.464
obs coarse	631	0.357	0.757	0.400	0.998	0.557	0.433
obs coarse	701	0.345	0.714	0.369	0.998	0.656	0.477
obs coarse	761	0.359	0.707	0.348	0.997	0.620	0.464
obs coarse	823	0.375	0.761	0.386	1.000	0.573	0.445
obs coarse	887	0.380	0.756	0.375	0.999	0.594	0.425
obs coarse	947	0.347	0.784	0.437	1.000	0.609	0.454
obs blind	11	0.358	0.400	0.042	1.000	0.568	0.442
obs blind	29	0.376	0.501	0.125	1.000	0.630	0.444
obs blind	47	0.342	0.423	0.081	1.000	0.651	0.454
obs blind	83	0.321	0.459	0.138	1.000	0.589	0.457
obs blind	131	0.340	0.463	0.124	1.000	0.615	0.426
obs blind	173	0.360	0.485	0.125	1.000	0.599	0.442
obs blind	211	0.355	0.519	0.164	1.000	0.615	0.450
obs blind	257	0.384	0.487	0.103	1.000	0.542	0.435
obs blind	307	0.336	0.417	0.081	1.000	0.615	0.441
obs blind	359	0.330	0.521	0.190	1.000	0.604	0.444

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
obs blind	401	0.360	0.456	0.096	1.000	0.620	0.437
obs blind	463	0.372	0.477	0.104	1.000	0.594	0.451
obs blind	509	0.297	0.486	0.189	1.000	0.641	0.436
obs blind	571	0.377	0.532	0.155	1.000	0.620	0.454
obs blind	631	0.349	0.491	0.143	1.000	0.562	0.426
obs blind	701	0.341	0.487	0.145	1.000	0.641	0.470
obs blind	761	0.352	0.486	0.134	1.000	0.630	0.468
obs blind	823	0.375	0.551	0.177	1.000	0.562	0.435
obs blind	887	0.374	0.500	0.126	1.000	0.604	0.423
obs blind	947	0.351	0.499	0.148	1.000	0.604	0.449
blind undercov.	11	0.518	0.460	-0.058	0.694	0.398	0.313
blind undercov.	29	0.472	0.496	0.024	0.697	0.414	0.324
blind undercov.	47	0.518	0.450	-0.068	0.683	0.434	0.330
blind undercov.	83	0.512	0.413	-0.099	0.689	0.430	0.357
blind undercov.	131	0.525	0.464	-0.061	0.681	0.434	0.315
blind undercov.	173	0.525	0.505	-0.020	0.680	0.469	0.316
blind undercov.	211	0.508	0.494	-0.015	0.670	0.461	0.327
blind undercov.	257	0.502	0.432	-0.070	0.699	0.406	0.309
blind undercov.	307	0.536	0.478	-0.057	0.697	0.477	0.339
blind undercov.	359	0.525	0.420	-0.105	0.706	0.426	0.315
blind undercov.	401	0.502	0.408	-0.094	0.704	0.398	0.334
blind undercov.	463	0.503	0.471	-0.031	0.692	0.391	0.352
blind undercov.	509	0.525	0.495	-0.030	0.705	0.434	0.348
blind undercov.	571	0.509	0.496	-0.014	0.692	0.418	0.362
blind undercov.	631	0.473	0.432	-0.040	0.683	0.418	0.297
blind undercov.	701	0.513	0.463	-0.050	0.686	0.430	0.323
blind undercov.	761	0.498	0.401	-0.098	0.698	0.438	0.316
blind undercov.	823	0.520	0.448	-0.072	0.703	0.449	0.332
blind undercov.	887	0.496	0.447	-0.049	0.706	0.426	0.343
blind undercov.	947	0.513	0.419	-0.094	0.688	0.449	0.322
sparse group04	11	0.282	0.832	0.551	0.994	0.486	0.407
sparse group04	29	0.290	0.848	0.558	0.997	0.528	0.442
sparse group04	47	0.301	0.829	0.528	0.991	0.472	0.413
sparse group04	83	0.246	0.853	0.607	0.998	0.458	0.432
sparse group04	131	0.320	0.838	0.518	0.996	0.479	0.433
sparse group04	173	0.309	0.880	0.571	0.994	0.472	0.399
sparse group04	211	0.286	0.852	0.566	0.993	0.493	0.404
sparse group04	257	0.274	0.858	0.584	0.996	0.431	0.419
sparse group04	307	0.303	0.857	0.555	0.991	0.403	0.412
sparse group04	359	0.244	0.820	0.575	0.992	0.493	0.412
sparse group04	401	0.278	0.817	0.539	0.988	0.479	0.417
sparse group04	463	0.305	0.870	0.564	0.996	0.507	0.435
sparse group04	509	0.296	0.864	0.568	0.986	0.424	0.439
sparse group04	571	0.251	0.836	0.586	0.996	0.472	0.408
sparse group04	631	0.287	0.845	0.558	0.993	0.451	0.399
sparse group04	701	0.262	0.835	0.573	0.998	0.472	0.435
sparse group04	761	0.286	0.781	0.495	0.987	0.493	0.429
sparse group04	823	0.259	0.908	0.649	0.996	0.500	0.410
sparse group04	887	0.287	0.861	0.574	0.994	0.479	0.412
sparse group04	947	0.289	0.829	0.540	0.998	0.500	0.385
sparse group08	11	0.337	0.924	0.587	0.999	0.497	0.409
sparse group08	29	0.296	0.897	0.601	1.000	0.455	0.434
sparse group08	47	0.324	0.894	0.570	0.995	0.462	0.425
sparse group08	83	0.302	0.915	0.613	0.996	0.455	0.417
sparse group08	131	0.307	0.931	0.624	0.998	0.458	0.430
sparse group08	173	0.312	0.911	0.599	0.995	0.469	0.415
sparse group08	211	0.299	0.921	0.622	0.999	0.455	0.405
sparse group08	257	0.318	0.901	0.582	0.999	0.462	0.414
sparse group08	307	0.302	0.912	0.610	0.998	0.417	0.419
sparse group08	359	0.324	0.894	0.570	0.995	0.472	0.419
sparse group08	401	0.296	0.885	0.589	0.996	0.479	0.417
sparse group08	463	0.302	0.931	0.629	0.999	0.507	0.429

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Table 10: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
sparse group08	509	0.318	0.923	0.605	0.994	0.476	0.428
sparse group08	571	0.302	0.916	0.614	0.999	0.427	0.415
sparse group08	631	0.321	0.904	0.583	0.997	0.465	0.417
sparse group08	701	0.316	0.903	0.587	0.999	0.490	0.434
sparse group08	761	0.298	0.914	0.616	0.997	0.479	0.434
sparse group08	823	0.311	0.903	0.591	0.996	0.444	0.402
sparse group08	887	0.306	0.916	0.610	0.997	0.476	0.419
sparse group08	947	0.309	0.911	0.602	0.998	0.465	0.416

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F Raw Error and Dispersion Rows

Table 11 records the calibrated-MSE side of the same matrix. These rows help distinguish two different claims: a method can have higher correlation with oracle credit while also changing calibrated error, zero-variance group frequency, or success-rate mix.

Table 11: Raw case-by-seed calibrated-error rows from the deep matrix.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
horizon 04 baseline	11	0.050	0.004	-0.046	0.000	0.290	0.422
horizon 04 baseline	29	0.053	0.003	-0.050	0.000	0.312	0.391
horizon 04 baseline	47	0.047	0.002	-0.045	0.000	0.286	0.396
horizon 04 baseline	83	0.044	0.003	-0.041	0.000	0.331	0.458
horizon 04 baseline	131	0.047	0.003	-0.044	0.000	0.297	0.391
horizon 04 baseline	173	0.049	0.004	-0.046	0.000	0.296	0.422
horizon 04 baseline	211	0.053	0.002	-0.051	0.000	0.310	0.422
horizon 04 baseline	257	0.054	0.003	-0.051	0.000	0.305	0.427
horizon 04 baseline	307	0.044	0.002	-0.042	0.000	0.317	0.438
horizon 04 baseline	359	0.058	0.003	-0.055	0.000	0.295	0.422
horizon 04 baseline	401	0.051	0.003	-0.048	0.000	0.298	0.438
horizon 04 baseline	463	0.051	0.003	-0.048	0.000	0.309	0.417
horizon 04 baseline	509	0.054	0.002	-0.052	0.000	0.294	0.385
horizon 04 baseline	571	0.052	0.004	-0.048	0.000	0.330	0.396
horizon 04 baseline	631	0.056	0.003	-0.053	0.000	0.265	0.474
horizon 04 baseline	701	0.052	0.004	-0.049	0.000	0.304	0.448
horizon 04 baseline	761	0.047	0.007	-0.040	0.000	0.299	0.443
horizon 04 baseline	823	0.045	0.003	-0.042	0.000	0.312	0.464
horizon 04 baseline	887	0.049	0.002	-0.047	0.000	0.301	0.422
horizon 04 baseline	947	0.050	0.002	-0.048	0.000	0.304	0.448
horizon 08 baseline	11	0.034	0.008	-0.026	0.000	0.361	0.542
horizon 08 baseline	29	0.036	0.007	-0.029	0.000	0.382	0.531
horizon 08 baseline	47	0.036	0.008	-0.028	0.000	0.395	0.542
horizon 08 baseline	83	0.036	0.006	-0.030	0.000	0.369	0.490
horizon 08 baseline	131	0.030	0.007	-0.022	0.000	0.362	0.552
horizon 08 baseline	173	0.038	0.007	-0.031	0.000	0.373	0.542
horizon 08 baseline	211	0.038	0.007	-0.031	0.000	0.347	0.547
horizon 08 baseline	257	0.035	0.006	-0.029	0.000	0.373	0.552
horizon 08 baseline	307	0.034	0.004	-0.030	0.000	0.364	0.583
horizon 08 baseline	359	0.039	0.004	-0.035	0.000	0.376	0.526
horizon 08 baseline	401	0.035	0.008	-0.027	0.000	0.334	0.516
horizon 08 baseline	463	0.037	0.006	-0.031	0.000	0.380	0.589
horizon 08 baseline	509	0.034	0.006	-0.028	0.000	0.369	0.599
horizon 08 baseline	571	0.040	0.003	-0.037	0.000	0.347	0.568
horizon 08 baseline	631	0.035	0.004	-0.031	0.000	0.341	0.510
horizon 08 baseline	701	0.033	0.006	-0.027	0.000	0.404	0.583
horizon 08 baseline	761	0.037	0.007	-0.030	0.000	0.350	0.547

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
horizon 08 baseline	823	0.031	0.006	-0.025	0.000	0.357	0.578
horizon 08 baseline	887	0.032	0.005	-0.028	0.000	0.364	0.604
horizon 08 baseline	947	0.037	0.005	-0.032	0.000	0.358	0.500
horizon 12 baseline	11	0.024	0.007	-0.017	0.000	0.415	0.609
horizon 12 baseline	29	0.028	0.005	-0.023	0.000	0.400	0.589
horizon 12 baseline	47	0.022	0.006	-0.016	0.000	0.417	0.677
horizon 12 baseline	83	0.027	0.007	-0.019	0.000	0.396	0.573
horizon 12 baseline	131	0.026	0.007	-0.019	0.000	0.397	0.604
horizon 12 baseline	173	0.026	0.009	-0.017	0.000	0.416	0.630
horizon 12 baseline	211	0.025	0.007	-0.018	0.000	0.390	0.667
horizon 12 baseline	257	0.023	0.005	-0.017	0.000	0.437	0.656
horizon 12 baseline	307	0.025	0.006	-0.018	0.000	0.412	0.667
horizon 12 baseline	359	0.024	0.008	-0.016	0.000	0.397	0.599
horizon 12 baseline	401	0.025	0.007	-0.018	0.000	0.412	0.630
horizon 12 baseline	463	0.022	0.007	-0.015	0.000	0.416	0.656
horizon 12 baseline	509	0.025	0.006	-0.019	0.000	0.425	0.661
horizon 12 baseline	571	0.024	0.008	-0.016	0.000	0.424	0.661
horizon 12 baseline	631	0.028	0.009	-0.019	0.000	0.392	0.604
horizon 12 baseline	701	0.024	0.005	-0.019	0.000	0.445	0.651
horizon 12 baseline	761	0.025	0.006	-0.019	0.000	0.405	0.625
horizon 12 baseline	823	0.027	0.009	-0.018	0.000	0.399	0.573
horizon 12 baseline	887	0.025	0.008	-0.017	0.000	0.402	0.646
horizon 12 baseline	947	0.025	0.008	-0.017	0.000	0.421	0.620
horizon 16 long wait	11	0.021	0.006	-0.015	0.000	0.513	0.667
horizon 16 long wait	29	0.018	0.004	-0.014	0.000	0.536	0.667
horizon 16 long wait	47	0.017	0.005	-0.013	0.000	0.538	0.718
horizon 16 long wait	83	0.020	0.005	-0.016	0.000	0.529	0.620
horizon 16 long wait	131	0.019	0.005	-0.014	0.000	0.542	0.671
horizon 16 long wait	173	0.019	0.005	-0.014	0.000	0.552	0.606
horizon 16 long wait	211	0.019	0.005	-0.014	0.000	0.546	0.718
horizon 16 long wait	257	0.017	0.005	-0.012	0.000	0.532	0.694
horizon 16 long wait	307	0.023	0.004	-0.019	0.000	0.539	0.671
horizon 16 long wait	359	0.016	0.005	-0.011	0.000	0.546	0.741
horizon 16 long wait	401	0.017	0.006	-0.011	0.000	0.539	0.718
horizon 16 long wait	463	0.018	0.005	-0.013	0.000	0.552	0.662
horizon 16 long wait	509	0.019	0.005	-0.014	0.000	0.545	0.662
horizon 16 long wait	571	0.018	0.006	-0.012	0.000	0.547	0.657
horizon 16 long wait	631	0.019	0.006	-0.013	0.000	0.517	0.667
horizon 16 long wait	701	0.017	0.005	-0.012	0.000	0.527	0.713
horizon 16 long wait	761	0.015	0.004	-0.011	0.000	0.562	0.736
horizon 16 long wait	823	0.022	0.006	-0.016	0.000	0.532	0.657
horizon 16 long wait	887	0.020	0.004	-0.016	0.000	0.539	0.713
horizon 16 long wait	947	0.018	0.005	-0.013	0.000	0.545	0.694
group 02 long wait	11	0.024	0.014	-0.010	0.031	0.507	0.641
group 02 long wait	29	0.024	0.010	-0.014	0.094	0.496	0.719
group 02 long wait	47	0.025	0.010	-0.015	0.062	0.495	0.656
group 02 long wait	83	0.022	0.011	-0.010	0.125	0.487	0.719
group 02 long wait	131	0.024	0.010	-0.013	0.094	0.514	0.625
group 02 long wait	173	0.024	0.009	-0.015	0.125	0.499	0.734
group 02 long wait	211	0.025	0.010	-0.014	0.031	0.489	0.578
group 02 long wait	257	0.024	0.010	-0.014	0.156	0.515	0.641
group 02 long wait	307	0.026	0.009	-0.017	0.062	0.533	0.672
group 02 long wait	359	0.026	0.014	-0.012	0.000	0.453	0.625
group 02 long wait	401	0.024	0.012	-0.012	0.062	0.528	0.672
group 02 long wait	463	0.022	0.008	-0.014	0.062	0.496	0.609
group 02 long wait	509	0.028	0.013	-0.015	0.125	0.515	0.656
group 02 long wait	571	0.035	0.009	-0.026	0.062	0.536	0.656
group 02 long wait	631	0.027	0.011	-0.016	0.000	0.492	0.641
group 02 long wait	701	0.028	0.013	-0.015	0.062	0.488	0.609
group 02 long wait	761	0.027	0.011	-0.016	0.031	0.490	0.672
group 02 long wait	823	0.024	0.009	-0.015	0.062	0.498	0.688
group 02 long wait	887	0.017	0.007	-0.010	0.031	0.537	0.734

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
group 02 long wait	947	0.024	0.008	-0.016	0.031	0.542	0.703
group 04 long wait	11	0.023	0.010	-0.014	0.000	0.488	0.680
group 04 long wait	29	0.022	0.006	-0.016	0.000	0.495	0.734
group 04 long wait	47	0.023	0.009	-0.014	0.000	0.481	0.656
group 04 long wait	83	0.022	0.006	-0.016	0.000	0.503	0.664
group 04 long wait	131	0.025	0.007	-0.018	0.000	0.524	0.625
group 04 long wait	173	0.025	0.010	-0.015	0.000	0.465	0.656
group 04 long wait	211	0.026	0.008	-0.018	0.000	0.507	0.609
group 04 long wait	257	0.021	0.006	-0.016	0.000	0.533	0.664
group 04 long wait	307	0.025	0.006	-0.019	0.000	0.495	0.641
group 04 long wait	359	0.029	0.008	-0.021	0.000	0.471	0.609
group 04 long wait	401	0.022	0.004	-0.018	0.000	0.544	0.656
group 04 long wait	463	0.027	0.007	-0.021	0.000	0.487	0.570
group 04 long wait	509	0.028	0.010	-0.018	0.000	0.516	0.617
group 04 long wait	571	0.026	0.007	-0.019	0.000	0.514	0.688
group 04 long wait	631	0.021	0.007	-0.014	0.000	0.485	0.680
group 04 long wait	701	0.023	0.006	-0.017	0.000	0.517	0.688
group 04 long wait	761	0.022	0.006	-0.017	0.000	0.491	0.688
group 04 long wait	823	0.024	0.007	-0.017	0.000	0.508	0.664
group 04 long wait	887	0.024	0.009	-0.015	0.000	0.508	0.672
group 04 long wait	947	0.020	0.007	-0.014	0.000	0.516	0.633
group 08 long wait	11	0.024	0.005	-0.019	0.000	0.506	0.656
group 08 long wait	29	0.026	0.005	-0.021	0.000	0.493	0.652
group 08 long wait	47	0.022	0.004	-0.018	0.000	0.506	0.680
group 08 long wait	83	0.025	0.004	-0.021	0.000	0.460	0.625
group 08 long wait	131	0.022	0.003	-0.019	0.000	0.522	0.637
group 08 long wait	173	0.025	0.007	-0.018	0.000	0.497	0.633
group 08 long wait	211	0.026	0.004	-0.021	0.000	0.503	0.621
group 08 long wait	257	0.025	0.003	-0.021	0.000	0.505	0.617
group 08 long wait	307	0.026	0.004	-0.021	0.000	0.511	0.664
group 08 long wait	359	0.024	0.006	-0.019	0.000	0.505	0.660
group 08 long wait	401	0.024	0.004	-0.020	0.000	0.515	0.660
group 08 long wait	463	0.025	0.005	-0.020	0.000	0.518	0.609
group 08 long wait	509	0.024	0.004	-0.020	0.000	0.503	0.691
group 08 long wait	571	0.023	0.007	-0.017	0.000	0.526	0.672
group 08 long wait	631	0.021	0.004	-0.017	0.000	0.517	0.609
group 08 long wait	701	0.021	0.005	-0.017	0.000	0.505	0.668
group 08 long wait	761	0.026	0.003	-0.023	0.000	0.494	0.586
group 08 long wait	823	0.024	0.006	-0.018	0.000	0.484	0.617
group 08 long wait	887	0.026	0.005	-0.021	0.000	0.484	0.613
group 08 long wait	947	0.020	0.004	-0.016	0.000	0.513	0.660
group 12 long wait	11	0.024	0.004	-0.020	0.000	0.496	0.646
group 12 long wait	29	0.026	0.004	-0.023	0.000	0.491	0.633
group 12 long wait	47	0.023	0.002	-0.021	0.000	0.511	0.659
group 12 long wait	83	0.023	0.003	-0.020	0.000	0.498	0.674
group 12 long wait	131	0.024	0.002	-0.022	0.000	0.531	0.638
group 12 long wait	173	0.023	0.003	-0.021	0.000	0.490	0.690
group 12 long wait	211	0.028	0.003	-0.024	0.000	0.488	0.617
group 12 long wait	257	0.025	0.003	-0.021	0.000	0.511	0.607
group 12 long wait	307	0.025	0.003	-0.022	0.000	0.502	0.654
group 12 long wait	359	0.024	0.003	-0.021	0.000	0.493	0.667
group 12 long wait	401	0.023	0.003	-0.020	0.000	0.502	0.674
group 12 long wait	463	0.024	0.004	-0.020	0.000	0.520	0.646
group 12 long wait	509	0.025	0.003	-0.022	0.000	0.490	0.635
group 12 long wait	571	0.024	0.004	-0.020	0.000	0.511	0.628
group 12 long wait	631	0.021	0.003	-0.018	0.000	0.507	0.656
group 12 long wait	701	0.021	0.003	-0.018	0.000	0.513	0.693
group 12 long wait	761	0.024	0.003	-0.020	0.000	0.500	0.659
group 12 long wait	823	0.026	0.003	-0.023	0.000	0.486	0.617
group 12 long wait	887	0.026	0.003	-0.023	0.000	0.496	0.635
group 12 long wait	947	0.023	0.003	-0.020	0.000	0.494	0.656
budget 002 full	11	0.029	0.027	-0.001	0.000	0.392	0.573

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
budget 002 full	29	0.028	0.028	-0.000	0.000	0.400	0.620
budget 002 full	47	0.029	0.029	0.000	0.000	0.385	0.609
budget 002 full	83	0.027	0.028	0.001	0.000	0.409	0.630
budget 002 full	131	0.028	0.028	-0.000	0.000	0.381	0.646
budget 002 full	173	0.036	0.033	-0.004	0.000	0.381	0.562
budget 002 full	211	0.028	0.029	0.001	0.000	0.398	0.630
budget 002 full	257	0.034	0.031	-0.003	0.000	0.375	0.516
budget 002 full	307	0.029	0.031	0.002	0.000	0.380	0.594
budget 002 full	359	0.035	0.032	-0.003	0.000	0.364	0.568
budget 002 full	401	0.030	0.032	0.002	0.000	0.374	0.573
budget 002 full	463	0.033	0.029	-0.003	0.000	0.407	0.557
budget 002 full	509	0.030	0.030	-0.001	0.000	0.381	0.604
budget 002 full	571	0.031	0.031	0.000	0.000	0.397	0.599
budget 002 full	631	0.031	0.032	0.001	0.000	0.388	0.562
budget 002 full	701	0.032	0.031	-0.001	0.000	0.406	0.594
budget 002 full	761	0.026	0.025	-0.001	0.000	0.413	0.651
budget 002 full	823	0.027	0.026	-0.001	0.000	0.392	0.615
budget 002 full	887	0.031	0.032	0.000	0.000	0.360	0.609
budget 002 full	947	0.033	0.032	-0.001	0.000	0.372	0.599
budget 008 full	11	0.029	0.026	-0.003	0.000	0.392	0.573
budget 008 full	29	0.028	0.024	-0.004	0.000	0.400	0.620
budget 008 full	47	0.029	0.024	-0.005	0.000	0.385	0.609
budget 008 full	83	0.027	0.024	-0.004	0.000	0.409	0.630
budget 008 full	131	0.028	0.026	-0.002	0.000	0.381	0.646
budget 008 full	173	0.036	0.027	-0.009	0.000	0.381	0.562
budget 008 full	211	0.028	0.025	-0.003	0.000	0.398	0.630
budget 008 full	257	0.034	0.028	-0.005	0.000	0.375	0.516
budget 008 full	307	0.029	0.028	-0.001	0.000	0.380	0.594
budget 008 full	359	0.035	0.028	-0.007	0.000	0.364	0.568
budget 008 full	401	0.030	0.027	-0.003	0.000	0.374	0.573
budget 008 full	463	0.033	0.030	-0.002	0.000	0.407	0.557
budget 008 full	509	0.030	0.028	-0.003	0.000	0.381	0.604
budget 008 full	571	0.031	0.028	-0.003	0.000	0.397	0.599
budget 008 full	631	0.031	0.031	0.000	0.000	0.388	0.562
budget 008 full	701	0.032	0.029	-0.003	0.000	0.406	0.594
budget 008 full	761	0.026	0.026	-0.000	0.000	0.413	0.651
budget 008 full	823	0.027	0.026	-0.001	0.000	0.392	0.615
budget 008 full	887	0.031	0.027	-0.004	0.000	0.360	0.609
budget 008 full	947	0.033	0.030	-0.003	0.000	0.372	0.599
budget 032 full	11	0.029	0.016	-0.013	0.000	0.392	0.573
budget 032 full	29	0.028	0.014	-0.014	0.000	0.400	0.620
budget 032 full	47	0.029	0.014	-0.014	0.000	0.385	0.609
budget 032 full	83	0.027	0.015	-0.012	0.000	0.409	0.630
budget 032 full	131	0.028	0.015	-0.013	0.000	0.381	0.646
budget 032 full	173	0.036	0.013	-0.023	0.000	0.381	0.562
budget 032 full	211	0.028	0.018	-0.010	0.000	0.398	0.630
budget 032 full	257	0.034	0.014	-0.019	0.000	0.375	0.516
budget 032 full	307	0.029	0.017	-0.013	0.000	0.380	0.594
budget 032 full	359	0.035	0.015	-0.020	0.000	0.364	0.568
budget 032 full	401	0.030	0.018	-0.012	0.000	0.374	0.573
budget 032 full	463	0.033	0.016	-0.016	0.000	0.407	0.557
budget 032 full	509	0.030	0.019	-0.012	0.000	0.381	0.604
budget 032 full	571	0.031	0.018	-0.013	0.000	0.397	0.599
budget 032 full	631	0.031	0.017	-0.014	0.000	0.388	0.562
budget 032 full	701	0.032	0.018	-0.014	0.000	0.406	0.594
budget 032 full	761	0.026	0.015	-0.011	0.000	0.413	0.651
budget 032 full	823	0.027	0.011	-0.015	0.000	0.392	0.615
budget 032 full	887	0.031	0.015	-0.017	0.000	0.360	0.609
budget 032 full	947	0.033	0.017	-0.016	0.000	0.372	0.599
budget 128 full	11	0.029	0.006	-0.023	0.000	0.392	0.573
budget 128 full	29	0.028	0.007	-0.021	0.000	0.400	0.620
budget 128 full	47	0.029	0.006	-0.023	0.000	0.385	0.609

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
budget 128 full	83	0.027	0.006	-0.021	0.000	0.409	0.630
budget 128 full	131	0.028	0.006	-0.022	0.000	0.381	0.646
budget 128 full	173	0.036	0.005	-0.032	0.000	0.381	0.562
budget 128 full	211	0.028	0.008	-0.020	0.000	0.398	0.630
budget 128 full	257	0.034	0.006	-0.027	0.000	0.375	0.516
budget 128 full	307	0.029	0.008	-0.022	0.000	0.380	0.594
budget 128 full	359	0.035	0.005	-0.030	0.000	0.364	0.568
budget 128 full	401	0.030	0.006	-0.024	0.000	0.374	0.573
budget 128 full	463	0.033	0.006	-0.027	0.000	0.407	0.557
budget 128 full	509	0.030	0.005	-0.025	0.000	0.381	0.604
budget 128 full	571	0.031	0.008	-0.023	0.000	0.397	0.599
budget 128 full	631	0.031	0.008	-0.023	0.000	0.388	0.562
budget 128 full	701	0.032	0.006	-0.026	0.000	0.406	0.594
budget 128 full	761	0.026	0.006	-0.020	0.000	0.413	0.651
budget 128 full	823	0.027	0.004	-0.023	0.000	0.392	0.615
budget 128 full	887	0.031	0.006	-0.025	0.000	0.360	0.609
budget 128 full	947	0.033	0.006	-0.027	0.000	0.372	0.599
obs full	11	0.029	0.010	-0.019	0.000	0.392	0.573
obs full	29	0.028	0.009	-0.019	0.000	0.400	0.620
obs full	47	0.029	0.008	-0.021	0.000	0.385	0.609
obs full	83	0.027	0.007	-0.020	0.000	0.409	0.630
obs full	131	0.028	0.009	-0.019	0.000	0.381	0.646
obs full	173	0.036	0.005	-0.031	0.000	0.381	0.562
obs full	211	0.028	0.010	-0.018	0.000	0.398	0.630
obs full	257	0.034	0.009	-0.024	0.000	0.375	0.516
obs full	307	0.029	0.008	-0.021	0.000	0.380	0.594
obs full	359	0.035	0.008	-0.027	0.000	0.364	0.568
obs full	401	0.030	0.012	-0.018	0.000	0.374	0.573
obs full	463	0.033	0.009	-0.024	0.000	0.407	0.557
obs full	509	0.030	0.007	-0.024	0.000	0.381	0.604
obs full	571	0.031	0.011	-0.020	0.000	0.397	0.599
obs full	631	0.031	0.010	-0.021	0.000	0.388	0.562
obs full	701	0.032	0.011	-0.021	0.000	0.406	0.594
obs full	761	0.026	0.008	-0.018	0.000	0.413	0.651
obs full	823	0.027	0.007	-0.020	0.000	0.392	0.615
obs full	887	0.031	0.007	-0.024	0.000	0.360	0.609
obs full	947	0.033	0.008	-0.024	0.000	0.372	0.599
obs coarse	11	0.029	0.017	-0.012	0.000	0.440	0.568
obs coarse	29	0.029	0.017	-0.012	0.000	0.453	0.630
obs coarse	47	0.025	0.014	-0.010	0.000	0.453	0.646
obs coarse	83	0.031	0.014	-0.017	0.000	0.456	0.578
obs coarse	131	0.031	0.017	-0.014	0.000	0.434	0.615
obs coarse	173	0.031	0.016	-0.014	0.000	0.446	0.609
obs coarse	211	0.028	0.015	-0.013	0.000	0.453	0.630
obs coarse	257	0.033	0.016	-0.017	0.000	0.441	0.547
obs coarse	307	0.028	0.016	-0.012	0.000	0.452	0.620
obs coarse	359	0.031	0.015	-0.016	0.000	0.461	0.599
obs coarse	401	0.029	0.014	-0.015	0.000	0.443	0.625
obs coarse	463	0.029	0.017	-0.012	0.000	0.457	0.604
obs coarse	509	0.030	0.015	-0.016	0.000	0.441	0.635
obs coarse	571	0.029	0.014	-0.015	0.000	0.464	0.625
obs coarse	631	0.033	0.016	-0.017	0.000	0.433	0.557
obs coarse	701	0.026	0.015	-0.012	0.000	0.477	0.656
obs coarse	761	0.028	0.016	-0.012	0.000	0.464	0.620
obs coarse	823	0.032	0.016	-0.016	0.000	0.445	0.573
obs coarse	887	0.033	0.017	-0.017	0.000	0.425	0.594
obs coarse	947	0.032	0.014	-0.018	0.000	0.454	0.609
obs blind	11	0.029	0.028	-0.001	0.000	0.442	0.568
obs blind	29	0.029	0.025	-0.004	0.000	0.444	0.630
obs blind	47	0.025	0.023	-0.002	0.000	0.454	0.651
obs blind	83	0.030	0.026	-0.004	0.000	0.457	0.589
obs blind	131	0.031	0.027	-0.003	0.000	0.426	0.615

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
obs blind	173	0.032	0.028	-0.004	0.000	0.442	0.599
obs blind	211	0.029	0.025	-0.005	0.000	0.450	0.615
obs blind	257	0.033	0.029	-0.003	0.000	0.435	0.542
obs blind	307	0.029	0.027	-0.002	0.000	0.441	0.615
obs blind	359	0.031	0.025	-0.006	0.000	0.444	0.604
obs blind	401	0.030	0.027	-0.003	0.000	0.437	0.620
obs blind	463	0.029	0.026	-0.003	0.000	0.451	0.594
obs blind	509	0.030	0.025	-0.005	0.000	0.436	0.641
obs blind	571	0.030	0.025	-0.005	0.000	0.454	0.620
obs blind	631	0.033	0.029	-0.005	0.000	0.426	0.562
obs blind	701	0.027	0.024	-0.004	0.000	0.470	0.641
obs blind	761	0.028	0.024	-0.004	0.000	0.468	0.630
obs blind	823	0.032	0.026	-0.006	0.000	0.435	0.562
obs blind	887	0.032	0.028	-0.004	0.000	0.423	0.604
obs blind	947	0.031	0.027	-0.004	0.000	0.449	0.604
blind undercov.	11	0.051	0.055	0.004	0.000	0.313	0.398
blind undercov.	29	0.053	0.052	-0.002	0.000	0.324	0.414
blind undercov.	47	0.048	0.052	0.004	0.000	0.330	0.434
blind undercov.	83	0.047	0.053	0.006	0.000	0.357	0.430
blind undercov.	131	0.049	0.054	0.004	0.000	0.315	0.434
blind undercov.	173	0.048	0.049	0.001	0.000	0.316	0.469
blind undercov.	211	0.049	0.050	0.001	0.000	0.327	0.461
blind undercov.	257	0.051	0.055	0.004	0.000	0.309	0.406
blind undercov.	307	0.043	0.046	0.003	0.000	0.339	0.477
blind undercov.	359	0.050	0.057	0.007	0.000	0.315	0.426
blind undercov.	401	0.047	0.053	0.005	0.000	0.334	0.398
blind undercov.	463	0.049	0.051	0.002	0.000	0.352	0.391
blind undercov.	509	0.048	0.050	0.002	0.000	0.348	0.434
blind undercov.	571	0.051	0.052	0.001	0.000	0.362	0.418
blind undercov.	631	0.053	0.055	0.002	0.000	0.297	0.418
blind undercov.	701	0.051	0.054	0.003	0.000	0.323	0.430
blind undercov.	761	0.051	0.057	0.006	0.000	0.316	0.438
blind undercov.	823	0.045	0.049	0.004	0.000	0.332	0.449
blind undercov.	887	0.048	0.051	0.003	0.000	0.343	0.426
blind undercov.	947	0.048	0.054	0.006	0.000	0.322	0.449
sparse group04	11	0.021	0.007	-0.014	0.000	0.407	0.486
sparse group04	29	0.023	0.007	-0.016	0.000	0.442	0.528
sparse group04	47	0.020	0.007	-0.013	0.000	0.413	0.472
sparse group04	83	0.024	0.007	-0.017	0.000	0.432	0.458
sparse group04	131	0.019	0.006	-0.013	0.000	0.433	0.479
sparse group04	173	0.024	0.006	-0.018	0.000	0.399	0.472
sparse group04	211	0.026	0.008	-0.018	0.000	0.404	0.493
sparse group04	257	0.025	0.007	-0.018	0.000	0.419	0.431
sparse group04	307	0.025	0.007	-0.018	0.000	0.412	0.403
sparse group04	359	0.022	0.008	-0.014	0.000	0.412	0.493
sparse group04	401	0.023	0.008	-0.014	0.000	0.417	0.479
sparse group04	463	0.020	0.005	-0.014	0.000	0.435	0.507
sparse group04	509	0.024	0.007	-0.017	0.000	0.439	0.424
sparse group04	571	0.027	0.009	-0.018	0.000	0.408	0.472
sparse group04	631	0.022	0.007	-0.015	0.000	0.399	0.451
sparse group04	701	0.023	0.007	-0.015	0.000	0.435	0.472
sparse group04	761	0.022	0.009	-0.013	0.000	0.429	0.493
sparse group04	823	0.026	0.005	-0.021	0.000	0.410	0.500
sparse group04	887	0.025	0.007	-0.018	0.000	0.412	0.479
sparse group04	947	0.022	0.008	-0.015	0.000	0.385	0.500
sparse group08	11	0.023	0.004	-0.019	0.000	0.409	0.497
sparse group08	29	0.024	0.005	-0.019	0.000	0.434	0.455
sparse group08	47	0.021	0.005	-0.016	0.000	0.425	0.462
sparse group08	83	0.022	0.004	-0.018	0.000	0.417	0.455
sparse group08	131	0.022	0.003	-0.019	0.000	0.430	0.458
sparse group08	173	0.022	0.004	-0.018	0.000	0.415	0.469
sparse group08	211	0.024	0.004	-0.020	0.000	0.405	0.455

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Table 11: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
sparse group08	257	0.023	0.005	-0.018	0.000	0.414	0.462
sparse group08	307	0.023	0.004	-0.019	0.000	0.419	0.417
sparse group08	359	0.022	0.005	-0.017	0.000	0.419	0.472
sparse group08	401	0.022	0.005	-0.017	0.000	0.417	0.479
sparse group08	463	0.022	0.003	-0.019	0.000	0.429	0.507
sparse group08	509	0.023	0.004	-0.019	0.000	0.428	0.476
sparse group08	571	0.024	0.004	-0.020	0.000	0.415	0.427
sparse group08	631	0.021	0.004	-0.017	0.000	0.417	0.465
sparse group08	701	0.021	0.004	-0.017	0.000	0.434	0.490
sparse group08	761	0.023	0.004	-0.019	0.000	0.434	0.479
sparse group08	823	0.024	0.005	-0.019	0.000	0.402	0.444
sparse group08	887	0.025	0.004	-0.021	0.000	0.419	0.476
sparse group08	947	0.020	0.004	-0.017	0.000	0.416	0.465

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G Reproducibility Checklist

1. Run the unit tests with `python3 -m unittest discover -s tests`.
2. Run the smoke gate with `./scripts/run_smoke.sh`.
3. Regenerate the deep matrix with `python3 -m experiments.deep_matrix`.
4. Regenerate LaTeX inputs with `python3 scripts/build_latex_inputs.py`.
5. Build the full paper with `./scripts/build_latex_paper.sh`.
6. Verify that the LaTeX artifact manifest records the PDF hash and page count.
7. Treat `public/ppo_grpo_opd_long_horizon.pdf` and the DOCX as abridged reports; treat the LaTeX PDF as the full paper artifact.

H Next Experiments

Closed-loop toy training. The next toy should train parameterized policies with a shared reward and compare closed-loop PPO, GRPO, and corrected GRPO variants under equal generated-token budgets. This will test whether estimator fidelity predicts policy improvement.

Process rewards. Adding process labels or subgoal rewards should narrow the gap between critic-based and group-relative estimators. This prediction is important because many practical systems do not rely on terminal reward alone.

Compaction. A compaction experiment should split long traces into variable numbers of sub-traces. The hypothesis is that critic-based individual-rollout objectives handle this naturally, while group-wise objectives need additional rules for matching sub-traces across completions.

Distillation hybrids. An OPD/OPSD follow-up should use successful RL traces as teacher or privileged self-teacher material and measure whether dense distillation preserves the credit assignment gains at lower inference cost.

I Replication Protocol for New Benchmarks

Trace schema. A real long-horizon benchmark should record more than a final reward. Each trajectory should include prompt identity, sampled response identity, token or step indices, tool calls, tool observations, compaction boundaries, verifier calls, process labels when available, terminal reward, and any safety or refusal events. This schema is the minimum needed to compare a group-relative estimator with a value-model estimator after the fact. Without it, the experiment can still report final success, but it cannot diagnose where credit was assigned or lost.

Budget matching. The clean comparison is not "PPO run" versus "GRPO run" in isolation. The comparison should state generated-token budget, number of prompts, completions per prompt, verifier calls, critic update steps, policy update steps, and peak activation or optimizer memory. A critic-based method should be charged for critic fitting and storage. A group-relative method should be charged for the extra completions needed to obtain reward contrast. A distillation method should be charged for teacher inference or self-teacher selection.

Primary measurements. For any benchmark with oracle or process annotations, report estimator correlation with oracle credit, calibrated MSE, sign accuracy, zero-variance group fraction, and return improvement under equal rollout budgets. For benchmarks without oracle credit, report held-out process-label agreement, per-step value calibration where labels exist, and sensitivity to trajectory length. Final score alone is not enough because a method can improve a short task while failing the mechanism required for long traces.

Decision rule. A public claim that one family is better should require at least three checks: the mean effect favors the method, the confidence interval excludes practical near-ties, and an adversarial or counterexample regime is reported. This paper's toy result follows that rule: it highlights the critic-favorable mean pattern, the near tie under tiny critic budget, and the group-favorable blind undercovered case. A larger benchmark should preserve the same discipline.

Audit package. The public release should include raw per-seed rows, aggregate tables, plotting scripts, exact command lines, environment notes, and a short statement of non-claims. The non-claims matter. For this workstream, the honest claim is that critic-style temporal information can dominate in observable long traces, while group-relative normalization remains strong when reward contrast is reliable and value modeling is weak. That is a research program, not a slogan.

J Artifact Provenance and Public Release Notes

Canonical data source. The canonical numeric artifact for this paper is the deep-matrix JSON under `results/`. The LaTeX macros, selected result table, axis summary, full case summary, raw seed rows, and calibrated-error appendix are generated from that JSON file by the LaTeX input builder. This keeps the manuscript from silently drifting away from the experiment output.

Rendered artifacts. The repository intentionally contains two kinds of public artifacts. The abridged PDF/DOCX report is designed for quick reading and sharing. The LaTeX PDF is the full paper artifact, including appendices and raw tables. The two artifacts should not be compared by page count; they serve different review contexts.

Manifest interpretation. The LaTeX manifest records input hashes, output hashes, the TeX engine, and the page-count gate. Because generated artifacts are committed with their manifest, the manifest's recorded git status necessarily describes the generation-time worktree rather than the final commit that contains the manifest. A clean checkout can rerun the build script to refresh the manifest and should recover the same result counts from the canonical JSON.

How to read the claim. The public headline should be the cautious one: 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample in this toy estimator benchmark. The stronger-sounding "17 mean critic wins" is true as a descriptive statistic, but the near-tie is scientifically important. The counterexample is equally important because it states a boundary condition for GRPO-style methods.

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